



MMS236 - Aircraft Design

Lecture 2

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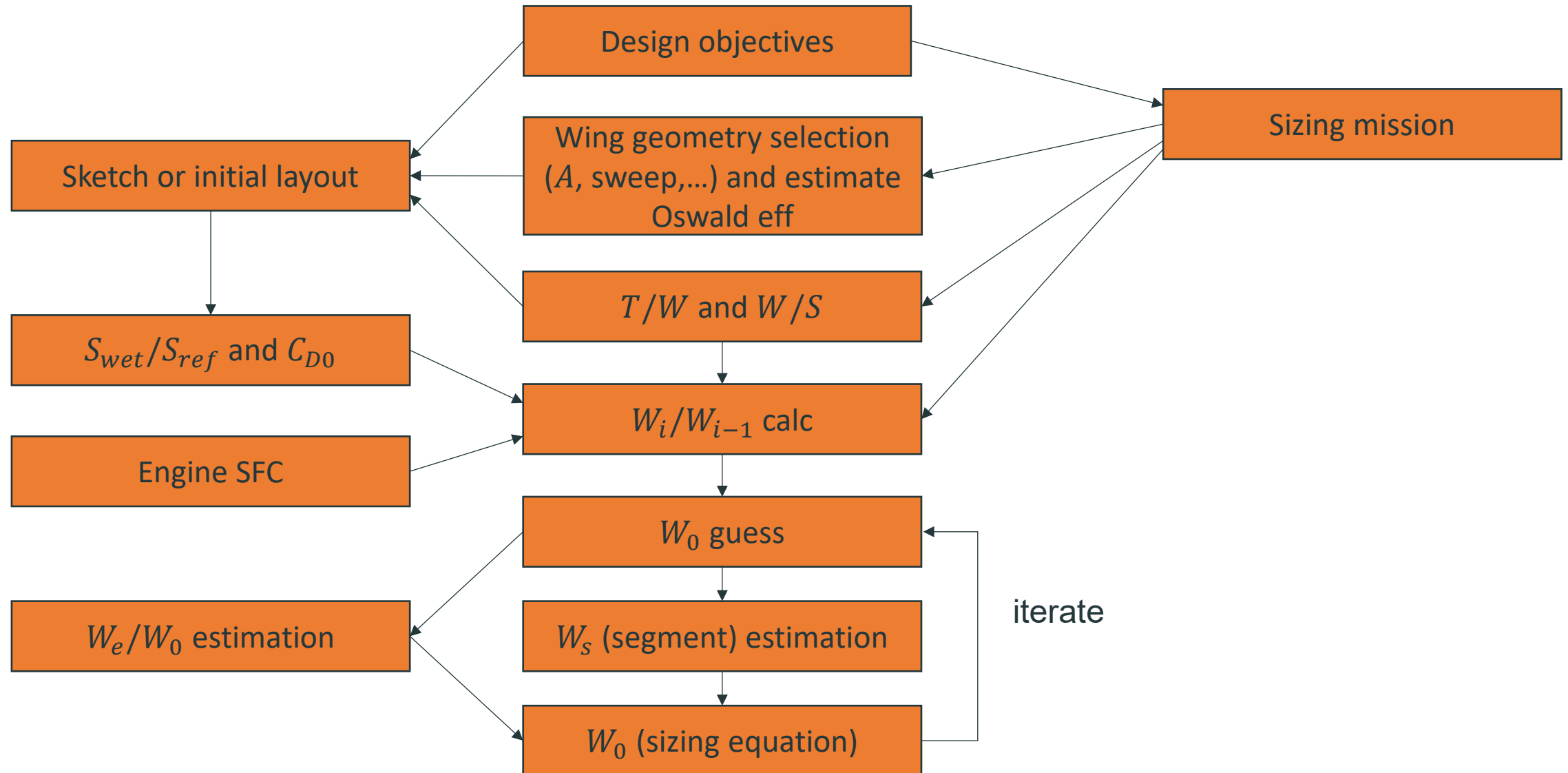
Division of Fluid Dynamics
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Initial sizing

Airfoil and wing/tail geometry section,
thrust-to-weight and wing loading

North American X – 15



Wing loading
 (W/S)

and

Thrust-to-weight
 (T/W)

Wing loading (W/S):

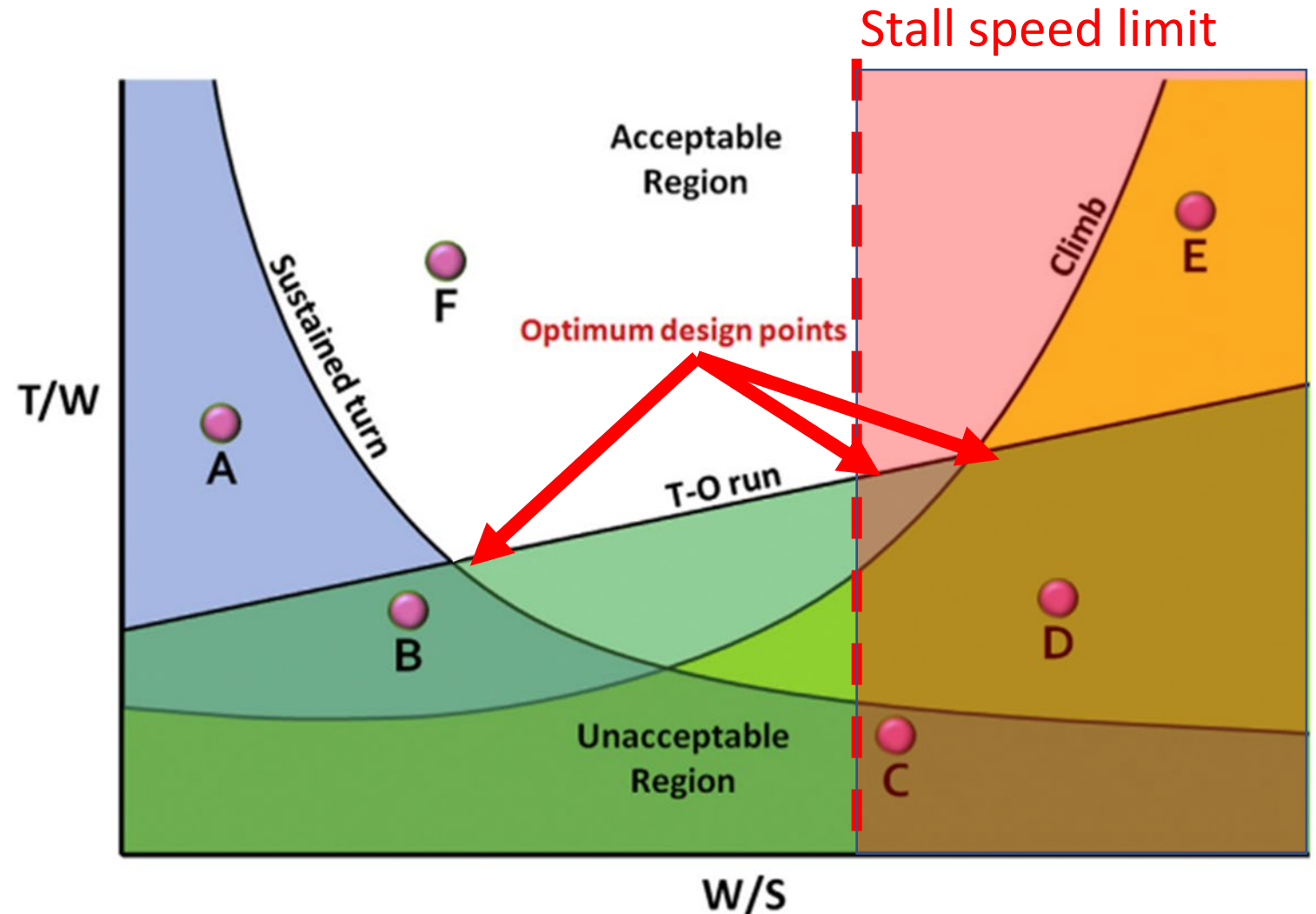
- ✈ Aircraft weight/mass divided by wing area.
- ✈ Set the size of the wing.
 - Low $W/S \Rightarrow$ Large wing
- ✈ Major impact on high-speed and low speed aerodynamic performance

Thrust to weight (T/W):

- ✈ Engine thrust divided by aircraft weight.
- ✈ Set the size of the engine.
 - High $T/W \Rightarrow$ Large engine
- ✈ Major impact on take-off and climb performance.
 - High $T/W \Rightarrow$ Shorter take-off & Landing
 - High $T/W \Rightarrow$ Higher climb rates
 - High $T/W \Rightarrow$ Higher initial cruise altitude

But increased engine weight.

- ✈ Allows to represent a feasible design space
- ✈ Values of T/W and W/S are provided at representative take-off conditions
- ✈ Typically, one favors the point that provides the minimum T/W
- ✈ If W/S is too low consider better flaps to increase $C_{L,max}$



Jet engine

	Typical Installed T/W
Jet fighter	0.6
Bomber/Cargo	0.25
Transport	0.25 – 0.4

$\frac{T}{W_0} = aM_{\max}^c$		
	a	c
Jet fighter	0.514	0.141
Bomber/Cargo	0.244	0.341
Transport	0.267	0.363

Prop: $\frac{P}{W} = \frac{Vg_0}{\eta_p} \frac{T}{W}$ (SI units)

Divide by P/W by $g_0 = 9.8 \text{ m/s}^2$ if P/W is given in W/kg.

	Typical Installed P/W (kW/kg)
General aviation single engine	0.12
Twin turboprop	0.33
Dash8-400	0.26
ATR42	0.17

$\frac{P}{W_0} = aV_{\max}^c$		
	a	c
General aviation single engine	0.036	0.22
Twin turboprop	0.016	0.5

✈ Use reference area (extend to center line).

- Low W/S => Lower stall speed
- Low W/S => Shorter take-off & Landing
- Low W/S => Improved maneuvering

But increased friction drag and weight.



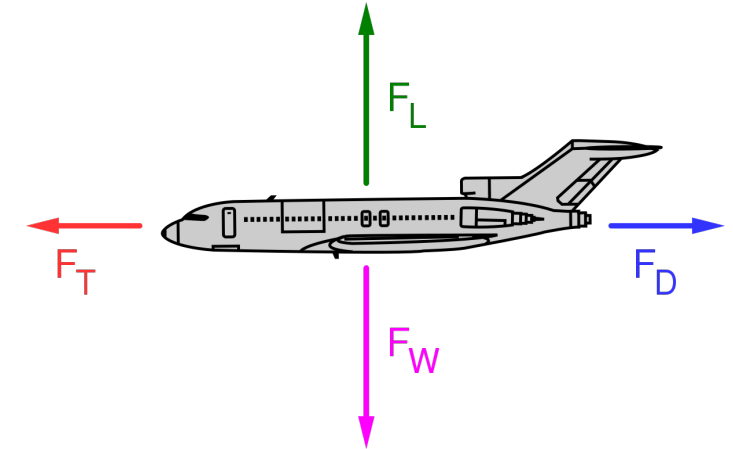
	Typical Installed W/S [kg/m ²]
Sailplane	30
General aviation single engine	83
Twin turboprop	195
Jet fighter	342
Transport	586 - 750
Q400 (twin turboprop)	453
777X	680
A350-1000	680
A321	750

✈ Thrust matching - cruise:

$$\left(\frac{T}{W}\right)_{cruise} = \frac{1}{\left(\frac{L}{D}\right)_{cruise}}$$

We know:

$$\text{✈ } \left(\frac{L}{D}\right)_{cruise} = \frac{C_L}{C_D} = \frac{C_L}{C_{D0} + KC_L^2} \quad \text{and} \quad C_L = \frac{L}{qS} = \frac{W}{qS}$$



V - Velocity

$\frac{L}{D}$ - Lift-to-Drag ratio

g_0 - gravitational constant 9.8 m/s^2

C_{D0} - Skin friction coef (zero lift Drag at cruise)

A - Aspect ratio

e - Oswald efficiency at cruise

q - Dynamic pressure ($\frac{1}{2}\rho V^2$)

W/S - Is the wing loading [kg/m^2]

$$\left(\frac{T}{W}\right)_{cruise} = \frac{qC_{D0}}{\left(\frac{W}{S}\right)_{cruise} \cdot g_0} + \left(\frac{W}{S}\right)_{cruise} \frac{g_0}{q\pi Ae}$$

✈ Thrust matching - climb:

$$\left(\frac{T}{W}\right)_{climb} = \frac{1}{\left(\frac{L}{D}\right)_{climb}} + \frac{V_{vertical}}{V}$$

V - Velocity

$\frac{L}{D}$ - Lift-to-Drag ratio

g_0 - gravitational constant 9.8 m/s²

C_{D0} - Skin friction coef (zero lift drag)

A - Aspect ratio

e - Oswald efficiency at climb

q - Dynamic pressure in climb ($\frac{1}{2}\rho V^2$)

W/S - Is the wing loading [kg/m²]

$$\left(\frac{T}{W}\right)_{climb} = \frac{q C_{D0}}{\left(\frac{W}{S}\right)_{climb} \cdot g_0} + \left(\frac{W}{S}\right)_{climb} \frac{g_0}{q \pi A e} + \frac{V_{vertical}}{V}$$

Typical climb rate ($V_{vertical}$) varies from aircraft type. Check the requirements

$$\left(\frac{T}{W}\right)_{take-off} = \left(\frac{T}{W}\right)_{cruise} \left(\frac{W_{cruise}}{W_{take-off}}\right) \left(\frac{T_{take-off}}{T_{cruise}}\right)$$

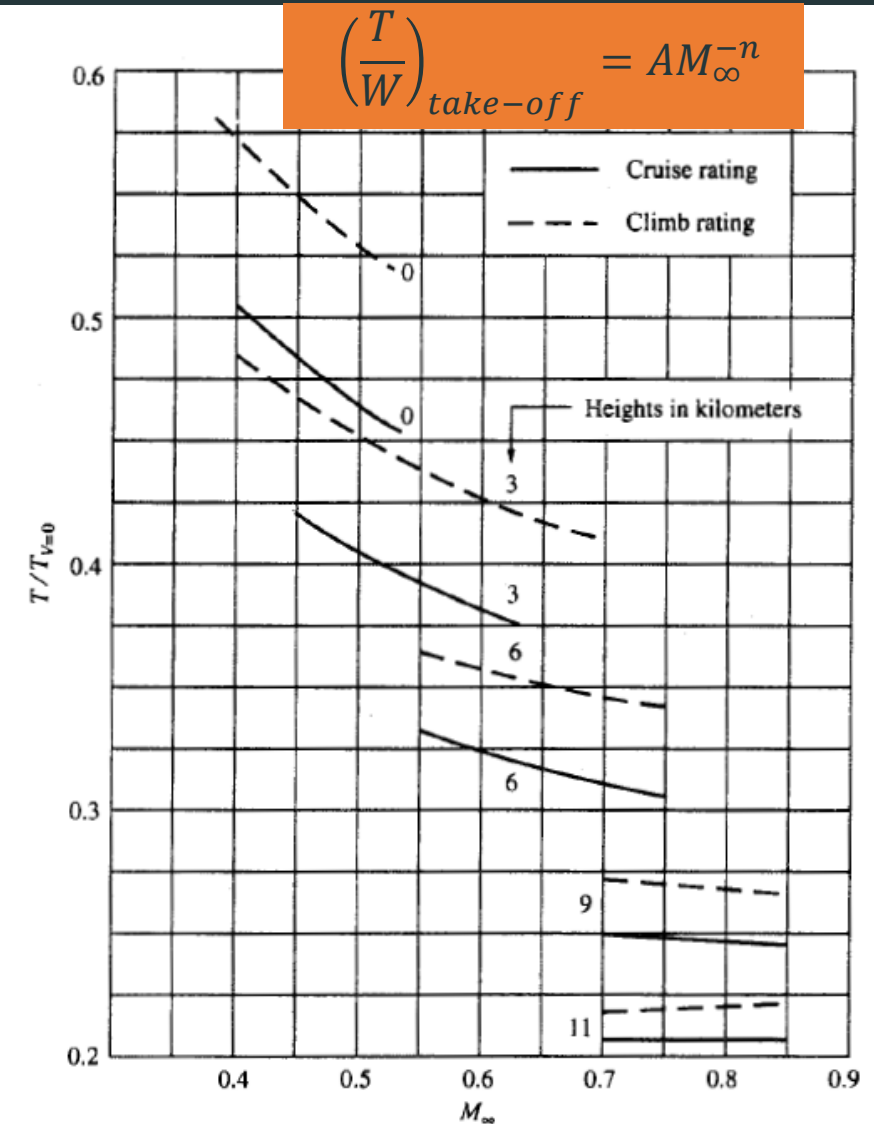
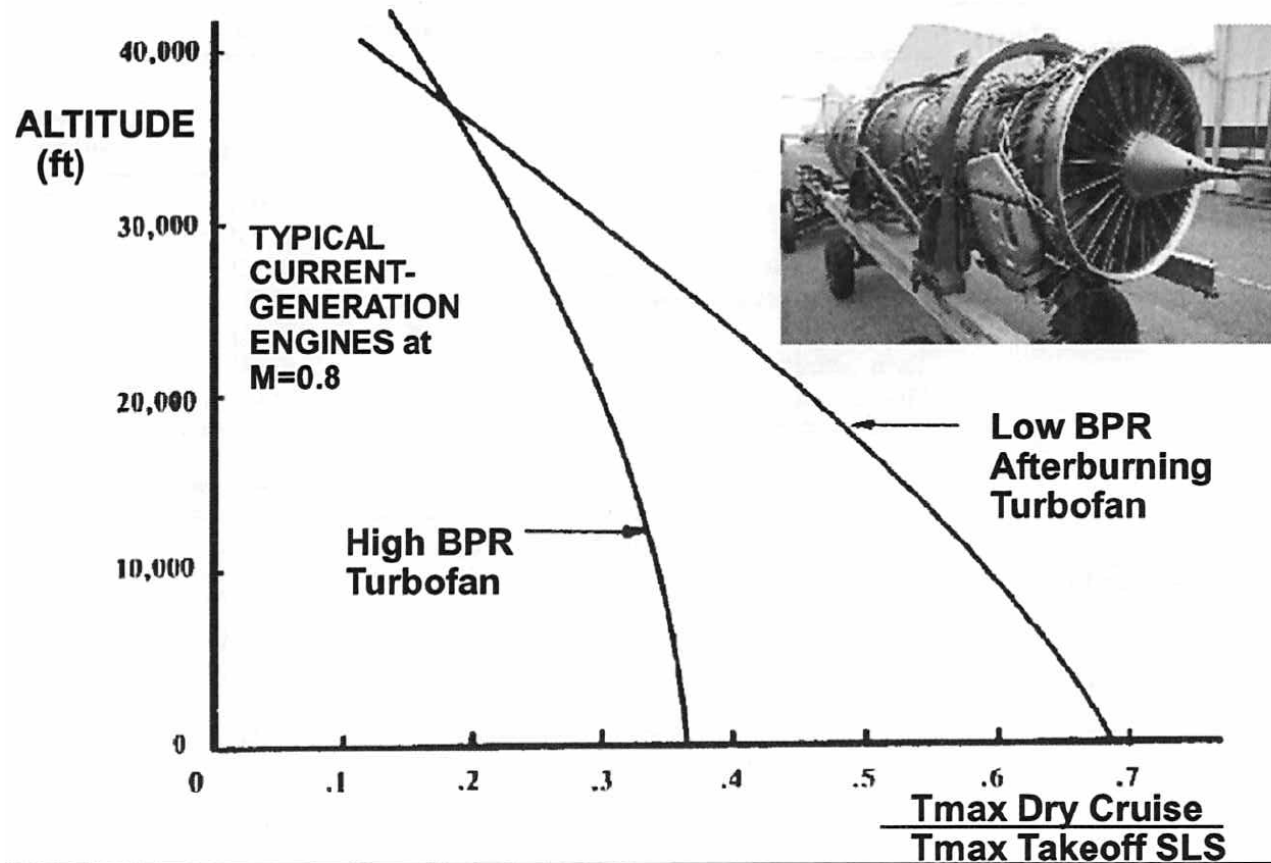


Figure 3.20

Variation of maximum thrust with Mach number and altitude for the Rolls-Royce RB211-535E4 turbofan. Note that $T_{V=0}$ is the thrust at zero velocity at sea level.

Turboprop:

- ✈ The core jet also contributes to thrust

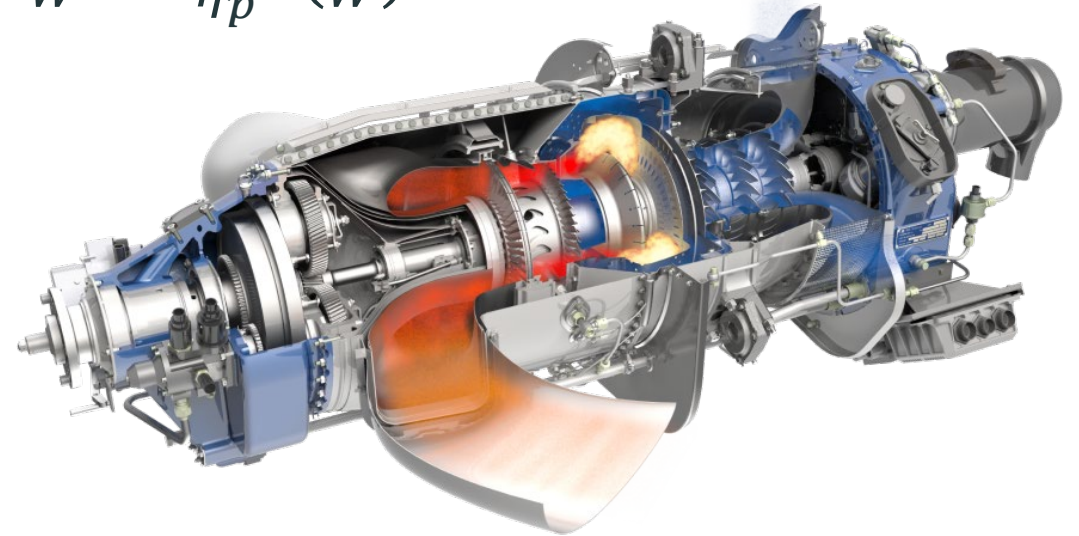
$$\left(\frac{P}{W}\right)_{take-off} = \left(\frac{P}{W}\right)_{cruise} \left(\frac{W_{cruise}}{W_{take-off}}\right) \left(\frac{P_{take-off}}{P_{cruise}}\right)$$

- ✈ The equivalent power (**eshp**) includes the shaft power and core jet thrust power

$$\left(\frac{T}{W}\right)_{cruise} = \frac{\eta_p}{V} \left(\frac{P}{W}\right)_{cruise} \frac{1}{g_0} \text{ (SI units)}$$

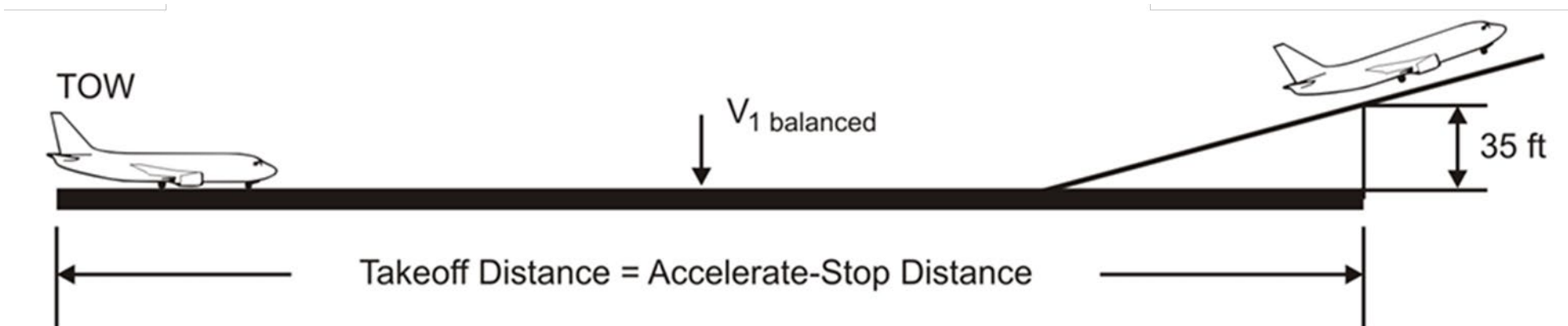
Prop: $\frac{P}{W} = \frac{V g_0}{\eta_p} \left(\frac{T}{W}\right) \text{ (SI units)}$

- ✈ Typical **eshp** at cruise between 60% -80% of take-off



- ✈ Required to meet a certain Take-off field length (usually provided).
- ✈ Take-off field length is the greater of:
 - 115% of the take-off distance with all engines operative, from break release to 35 ft obstacle clearance.
 - Balance field length: distance required to take-off and clear 35ft obstacle with one engine inoperative (OEI) at the worst possible time (reach decision speed).

Decision speed V_1 defined for: Accelerate-stop distance = take-off distance OEI



- ✦ Both T/W and W/L play contribute take-off distance.
- ✦ Can be determined with the take-off parameter (TOP).

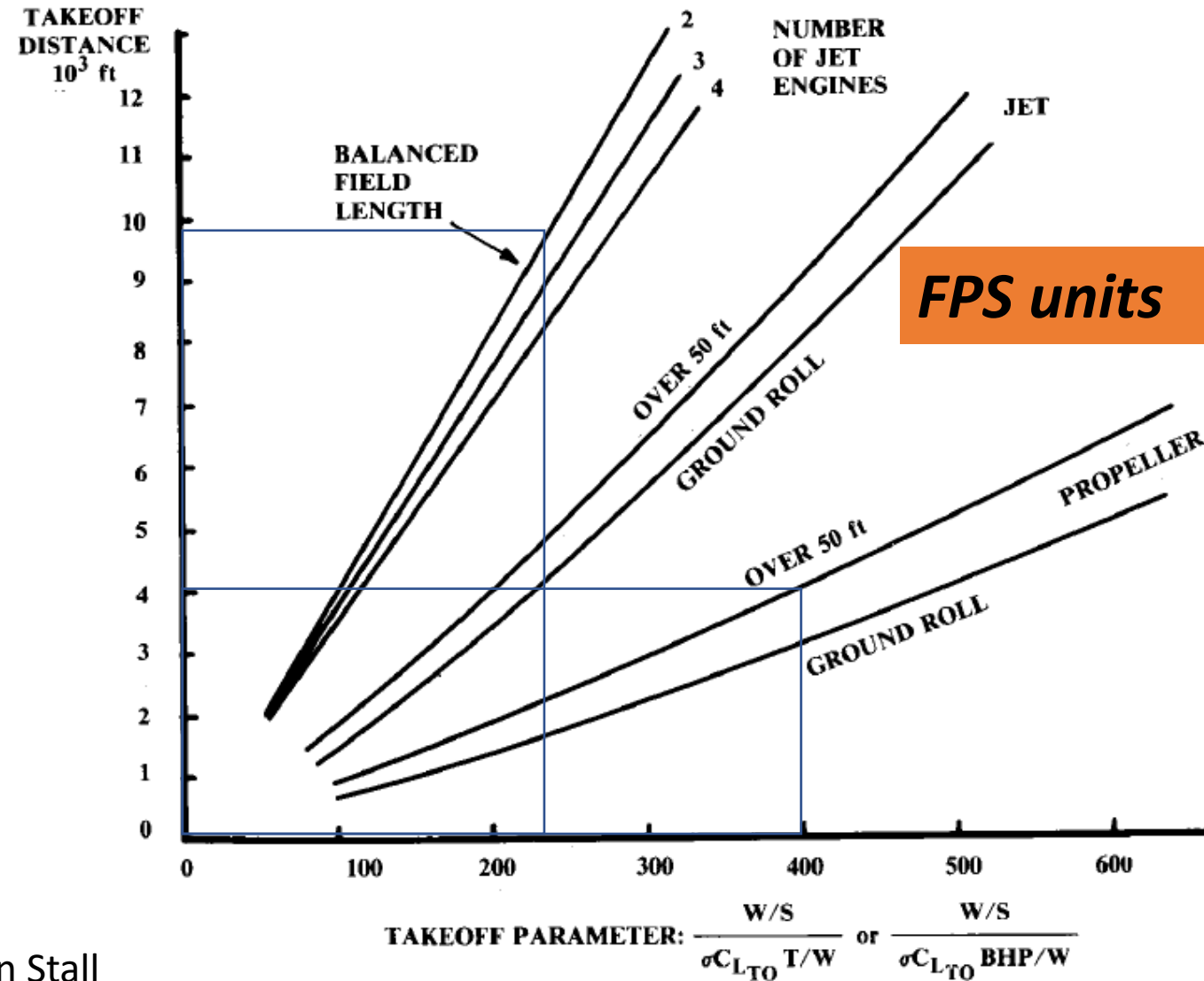
$$\frac{W}{S} = \text{TOP} \sigma C_{L,TO} \left(\frac{T}{W} \right) \quad (\text{jet})$$

$$\frac{W}{S} = \text{TOP} \sigma C_{L,TO} \left(\frac{hp}{W} \right) \quad (\text{prop})$$

$\sigma = \frac{\rho}{\rho_{SL}}$ density ratio:

$C_{L,TO}$ – Lift coefficient at take-off ($C_{L_{max}}/1.21$)

take-off speed is 10% higher than Stall



✦ Can also be determined for ground roll when provided, approximated as:

$$S_g = \frac{1}{2gK_A} \ln \left(1 + \frac{K_A}{K_T} V_{LOF}^2 \right) + NV_{LOF}$$

where:

$$K_T = \frac{T_{av}}{W} - \mu, \quad T_{av} \text{ is the average thrust during take-off (use a thrust value at about } \frac{1}{\sqrt{2}} V_{LOF})$$

$$K_A = \frac{\rho}{2 \left(\frac{W}{S} \right)} \left(\mu C_{L,TO \text{ run}} - C_{D,TO \text{ run}} \right)$$

S_g - Ground roll [m], distance from break release from wheels leaving the ground

N – Time for rotation (s), typically 3 for a large aircraft

$C_{L,TO \text{ run}}$ – Average Lift coefficient during take-off (**before rotation**)

μ – ground friction constant

V_{LOF} – liftoff speed [m/s]

$C_{D,TO \text{ run}}$ – Average total drag coefficient at take-off

(**includes C_{d0} and C_{di} , note that C_{di} is reduced due to ground effect**)

W/S – Is the wing loading [kg/m²]

$$\frac{T}{W} = \frac{K_A V_{LOF}^2}{\exp \left((S_g - NV_{LOF}) 2gK_A \right) - 1} + \mu$$

✈ Wing loading for stall

$$W = L = q_{stall} \cdot S \cdot C_{L,max} = \frac{1}{2g_0} \rho V_{stall}^2 \cdot S \cdot C_{L,max}$$

Hence:

$$\left(\frac{W}{S}\right)_{stall} = \frac{1}{2g_0} \rho V_{stall}^2 \cdot C_{L,max} \dots \left[\frac{\text{kg}}{\text{m}^2}\right]$$

$C_{L,max}$ - maximum lift coefficient

V_{stall} - Stall velocity [m/s]

ρ - Density [kg/m³] (typically sea level for stall calculations)

g_0 - gravitational constant 9.8 m/s²



- ✈ V_{stall} determined from regulations.
 - E.g. FAR23: for certain aircraft with $< 12,500\text{lb}$ TOGW – $V_{stall} < 61$ kts.

- ✈ It can also be determined from the approach speed:

$$V_{app} \geq 1.23V_{stall}$$

(for civil transport applications FAR25)

$$V_{app} \geq 1.3V_{stall}$$

(for civil transport applications FAR23)

- Approach speed is normally stated in requirements.
- ✈ Look for similar aircraft to find out!

- ✈ $C_{L,max}$ changes with:
 - Airfoil
 - Wing geometry
 - Flap and slat geometries
 - Reynolds number
 - Surface quality
 -

Requires extensive analysis, for now we rely on simple methods!

$$1.2 < C_{L,max} < 1.5$$

(for plain wing)

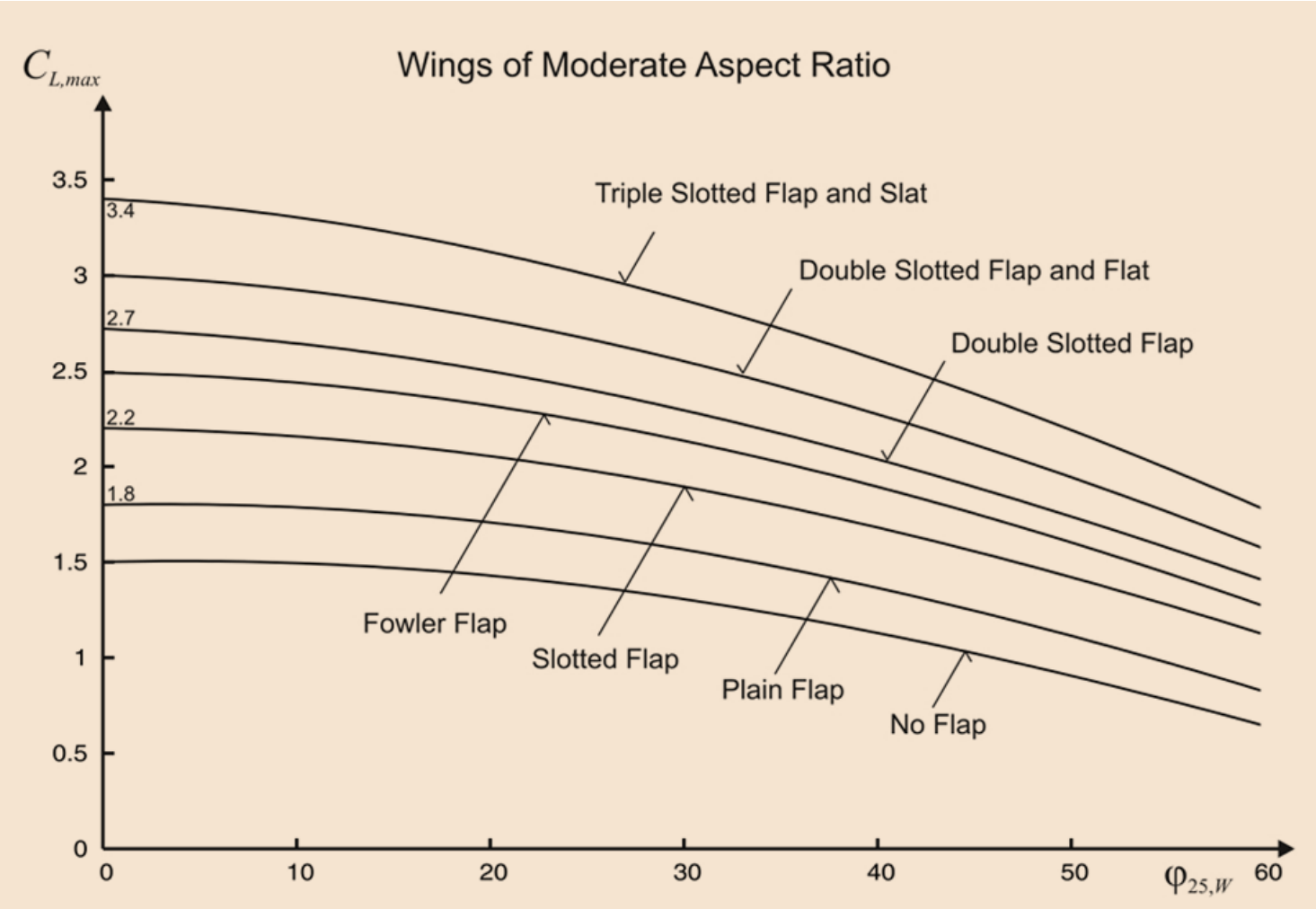
$$C_{L,max} = 0.9C_\ell \cos(\Lambda_{0.25c})$$

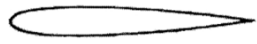
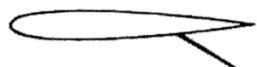
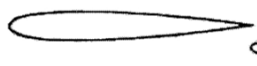
(for $AR > 5$)

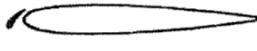
C_ℓ - Airfoil max lift coefficient

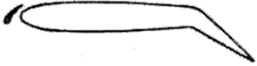
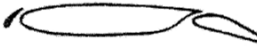
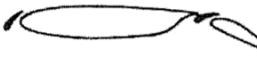
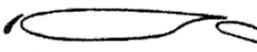
$\Lambda_{0.25c}$ - Sweep at $\frac{1}{4}$ chord

✈ Max Lift Coefficient

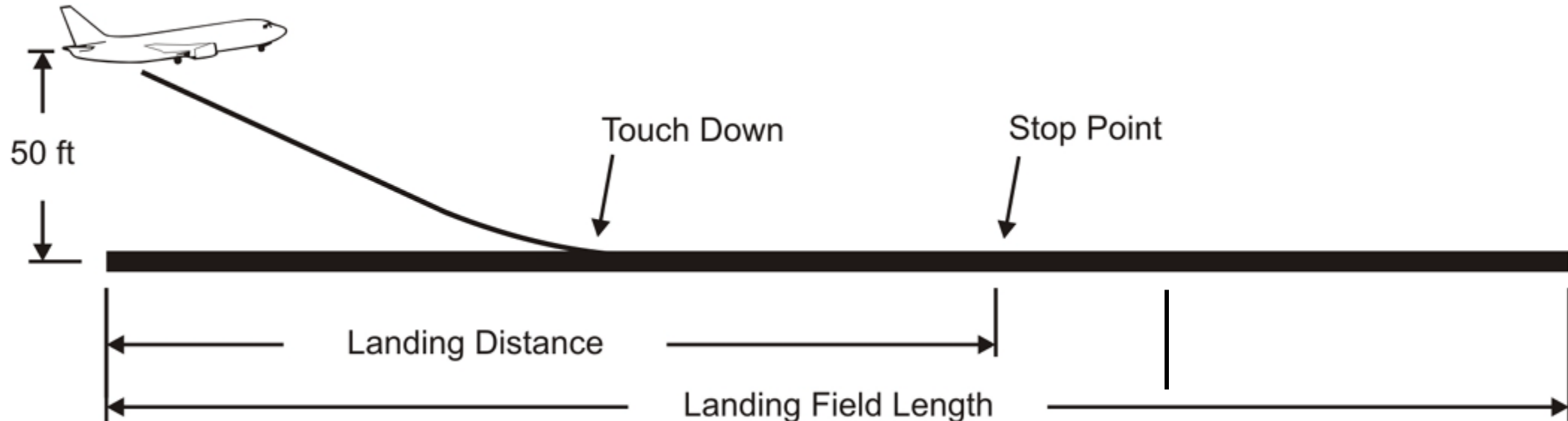


		$C_{L,max}$	$\Delta C_{L,max}$
Clean Airfoil		1,45	-
Plain Flap		2,25	0,80
Single-Slotted Flap		2,60	1,15
Double-Slotted Flap		2,80	1,35
Split Flap		2,40	0,95
Double-Wing (Junkers)		2,25	0,80

		$C_{L,max}$	$\Delta C_{L,max}$
Fowler Flap		2,80	1,35
Slat		2,00	0,55

Combinations:		$C_{L,max}$	$\Delta C_{L,max}$
Plain Flap and Slat		2,45	1,00
Single-Slotted Flap and Slat		2,70	1,25
Double-Slotted Flap and Slat		2,90	1,45
Fowler Flap and Slat		3,00	1,55

- ✈ Required to meet a certain landing field length (usually provided).
- ✈ Landing field length is the:
 - Horizontal distance to land and stop including a 50 ft (15.24 m) obstacle clearance at approach. – **Applicable up and including FAR23**
 - **FAR25:** Dry Runways Regulation adds an arbitrary margin to the total distance
 - A turbofan must land within **60%** of the available distance
 - A turboprop must land within **70%** of the available distance



- ✈ Only depends on wing loading.

$$S_{landing} = 4.84 \frac{W}{S} \left(\frac{1}{\sigma C_{L,max}} \right) + S_a$$

$C_{L,max}$ – Maximum lift coefficient

$S_{landing}$ – Landing distance [m]

S_a – Obstacle clearance distance [m], 305 m for airliner, 183 m for general aviation, 137 m for STOL

W/S – Landing wing-loading [kg/m²]

$\sigma = \frac{\rho}{\rho_{SL}}$ density ratio:

- ✈ Multiply by 1.67 (turbofan) or 1.43 (turboprop) to get the FAR25 landing field length.

- ✈ W/S Must be converted to take-off conditions for comparison

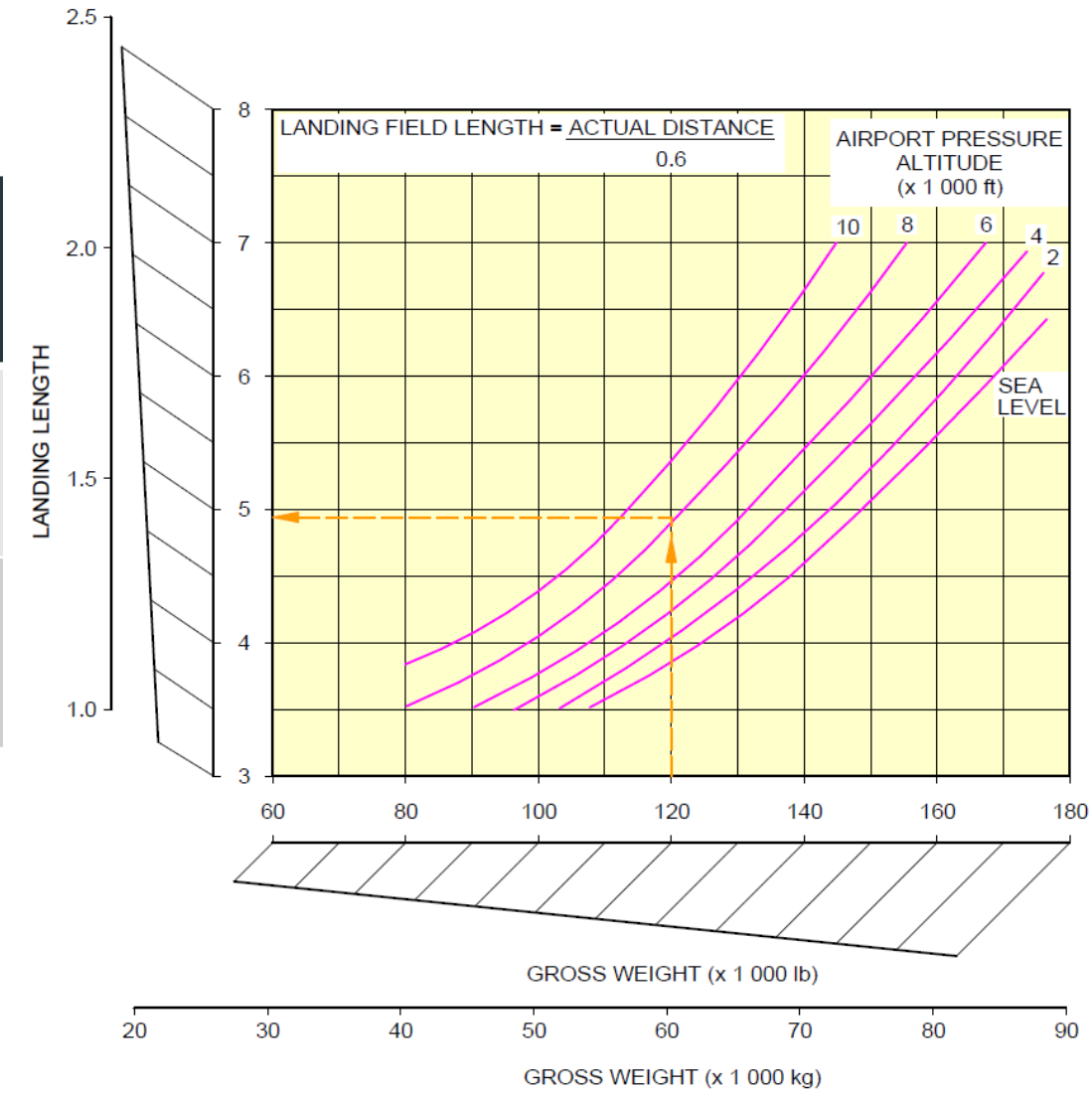
$$\left(\frac{W}{S} \right)_{T/O} = \frac{\left(\frac{W}{S} \right)_{landing}}{\frac{W_{landing}}{W_{T/O}}}$$

Runway lengths of selected aircraft

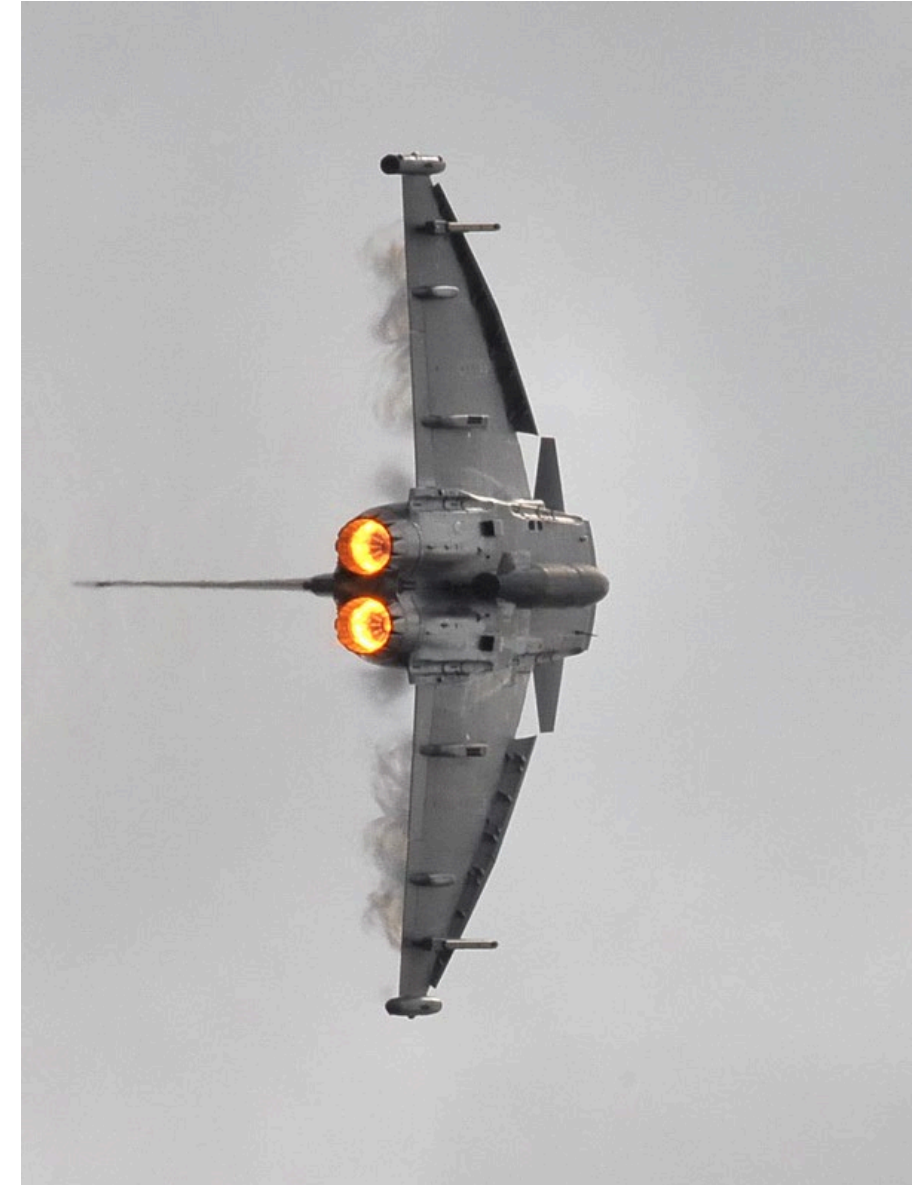
CHALMERS

Aircraft	Q400	Airbus A320-200	Boeing 737-600	Cessna 525	Boeing 747
TOFL [m]	1400	2336			
LFL [m]	1350	1470			

Keep an eye on these when you design similar aircraft, ACAP documents are a good source for field performance data



- ✈ Can be a limiting parameter for certain configurations
- ✈ Instantaneous turn.
 - Maximum value set by aerodynamic (stall) or mechanical (structural) limitations.
- ✈ Sustained turn
 - The thrust is sufficient to maintain velocity and altitude during the turn.



- ✈ Load factor (n) or g-loading is the acceleration due to lift.
 - Expressed as a multiple of $g_0 = 9.8 \text{ m/s}^2$
 - Limit load factor up to 9g in modern aircraft (**note**: depends on weight)

$$\left(\frac{W}{S}\right)_{turn} = \frac{q C_L}{g_0 n} \left[\frac{\text{kg}}{\text{m}^2} \right]$$

- ✈ Turn rate, $\dot{\psi}$ is the radial acceleration divided by the velocity.
 - This is usually specified for some particular flight conditions, use it to calculate **n** .

$$\dot{\psi} = \frac{g \sqrt{n^2 - 1}}{V} \left[\frac{\text{rad}}{\text{s}} \right]$$

q – Dynamic pressure [Pa]

V – Velocity [m/s]

C_L – Lift coefficient

$$\left(\frac{W}{S}\right)_{T/O} = \frac{\left(\frac{W}{S}\right)_{turn}}{\frac{W_{turn}}{W_{T/O}}}$$

✈ A maximum load factor (n) is normally specified at a given flight condition.

- E.g. 5g at M=0.9 30,000 ft
- Without slowing down, speed is maintained, hence thrust = drag.
- Lift = Weight*load factor

$$\left(\frac{W}{S}\right)_{sust.turn} = \frac{\left(\frac{T}{W}\right)_{turn} \pm \sqrt{\left(\frac{T}{W}\right)_{turn}^2 - (4n^2 C_{D0} / [\pi A e])}}{g_0 2n^2 / (q \pi A e)} \left[\frac{\text{kg}}{\text{m}^2} \right]$$

Note: if values of W/S deviate too much from historical data check e .

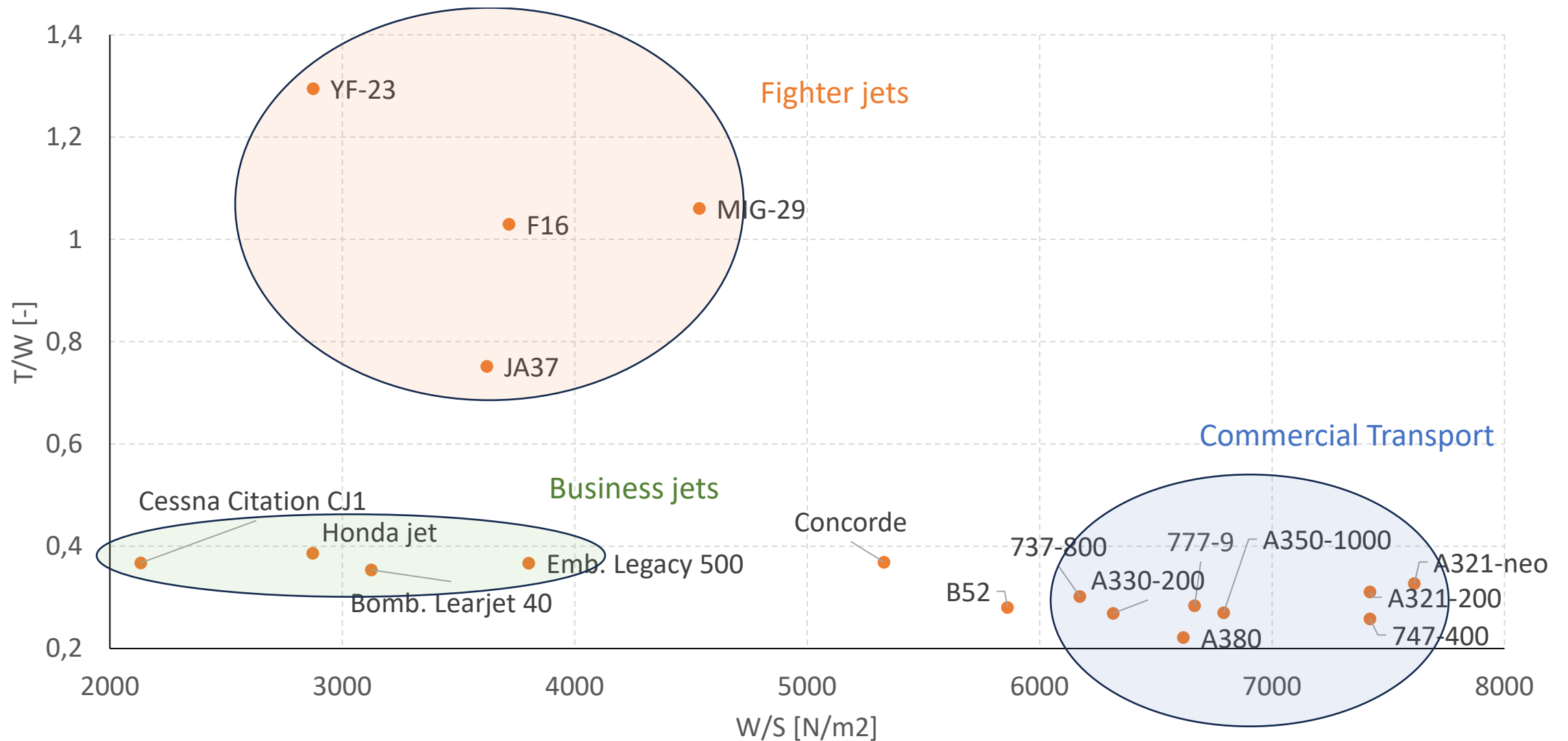
A – Aspect ratio

C_{D0} – Zero lift drag coefficient

e - Oswald efficiency, varies with C_L and reduces for high C_L

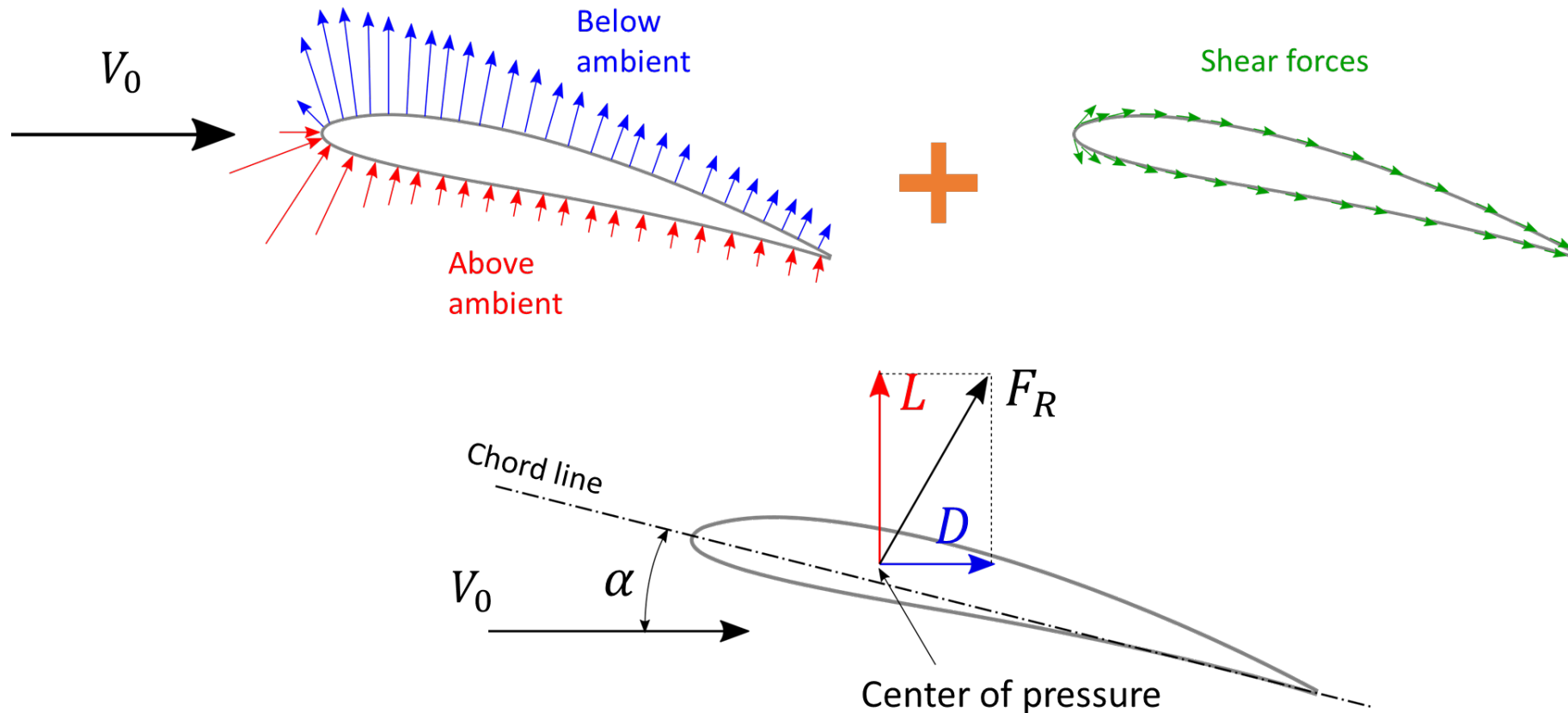
q – Dynamic pressure [Pa]

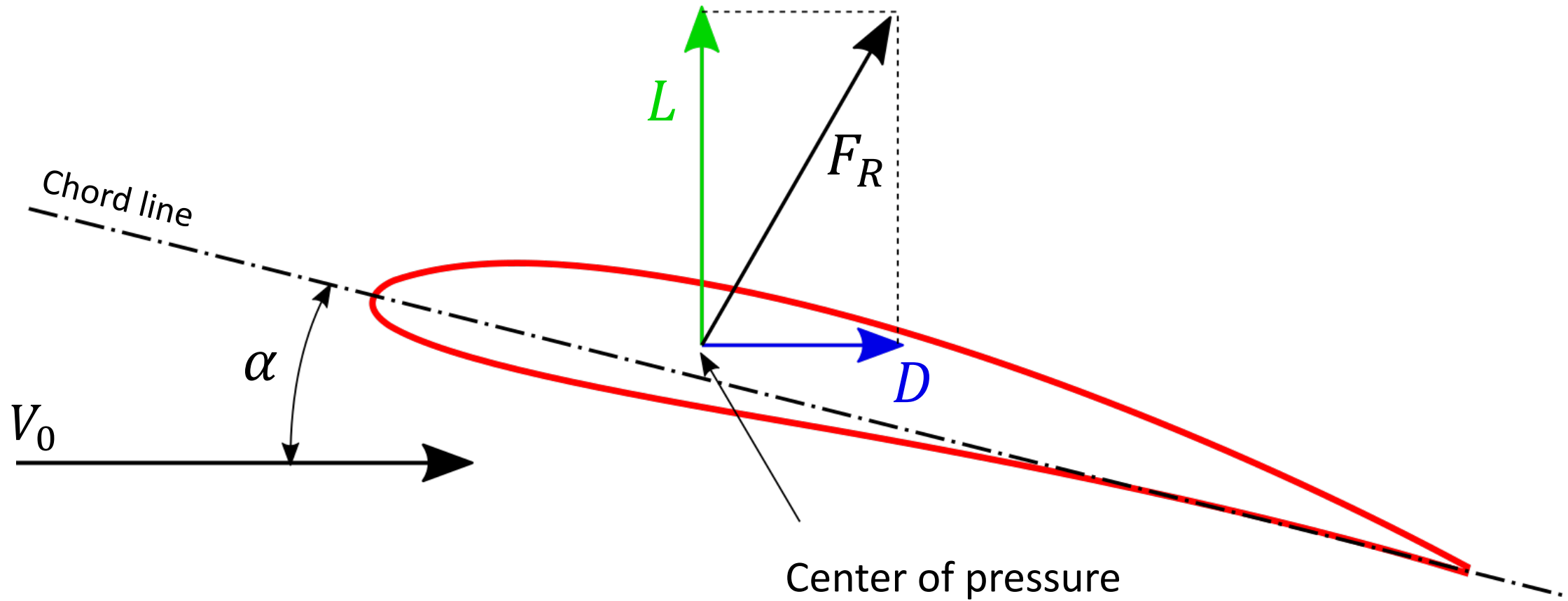
$$\left(\frac{W}{S}\right)_{T/O} = \frac{\left(\frac{W}{S}\right)_{turn}}{\frac{W_{turn}}{W_{T/O}}}$$



Selection of wing and tail geometry

- ✈ **Bernoulli theorem:** Lift is generated due to a pressure difference in the upper and lower side of the airfoil.





✈ Section lift coefficient:

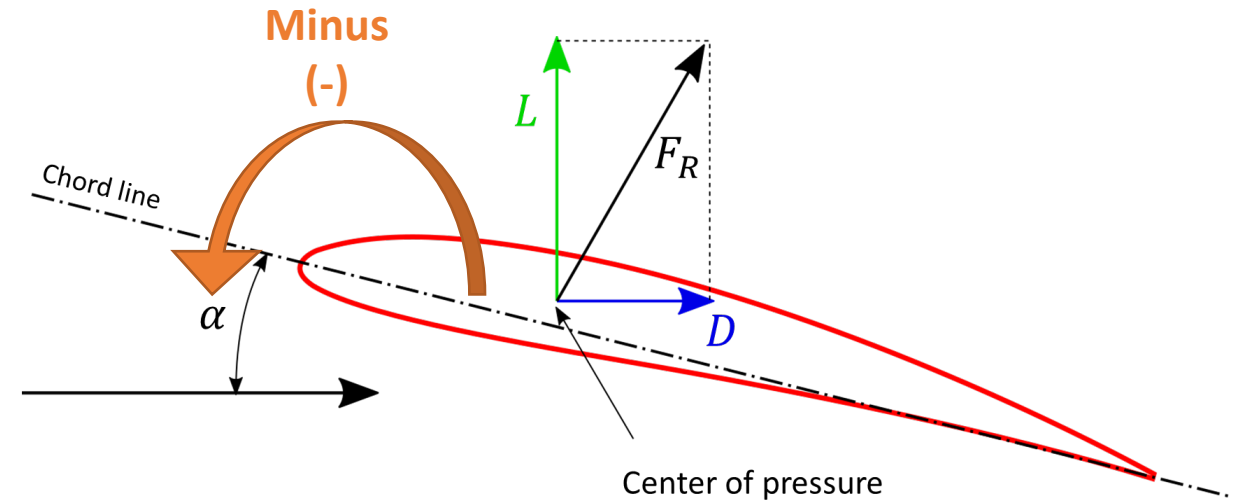
$$C_l = \frac{L'}{qc}$$

✈ Section drag coefficient:

$$C_d = \frac{D'}{qc}$$

✈ Section pitching moment coefficient:

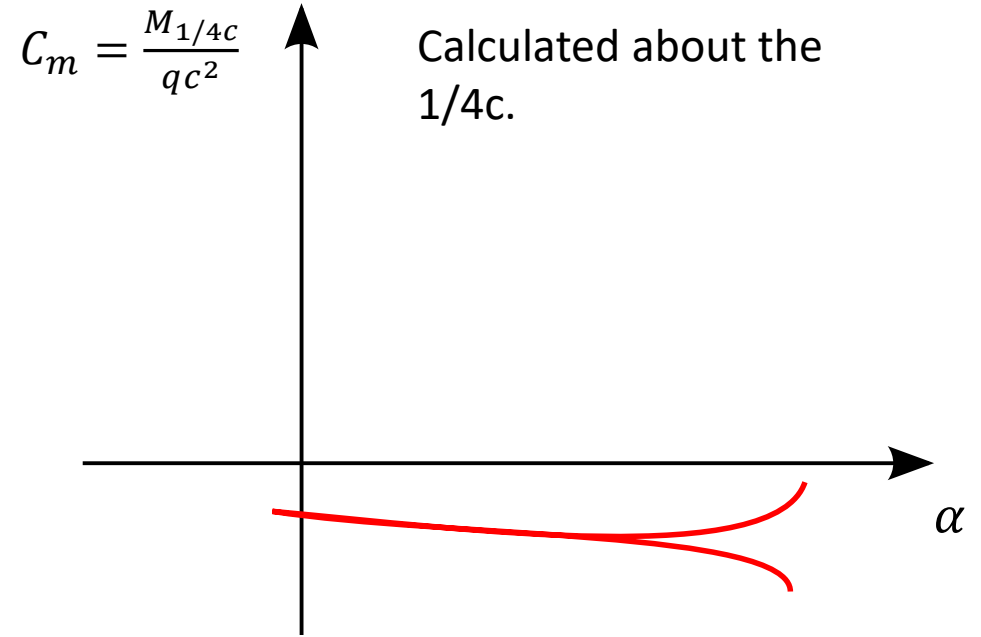
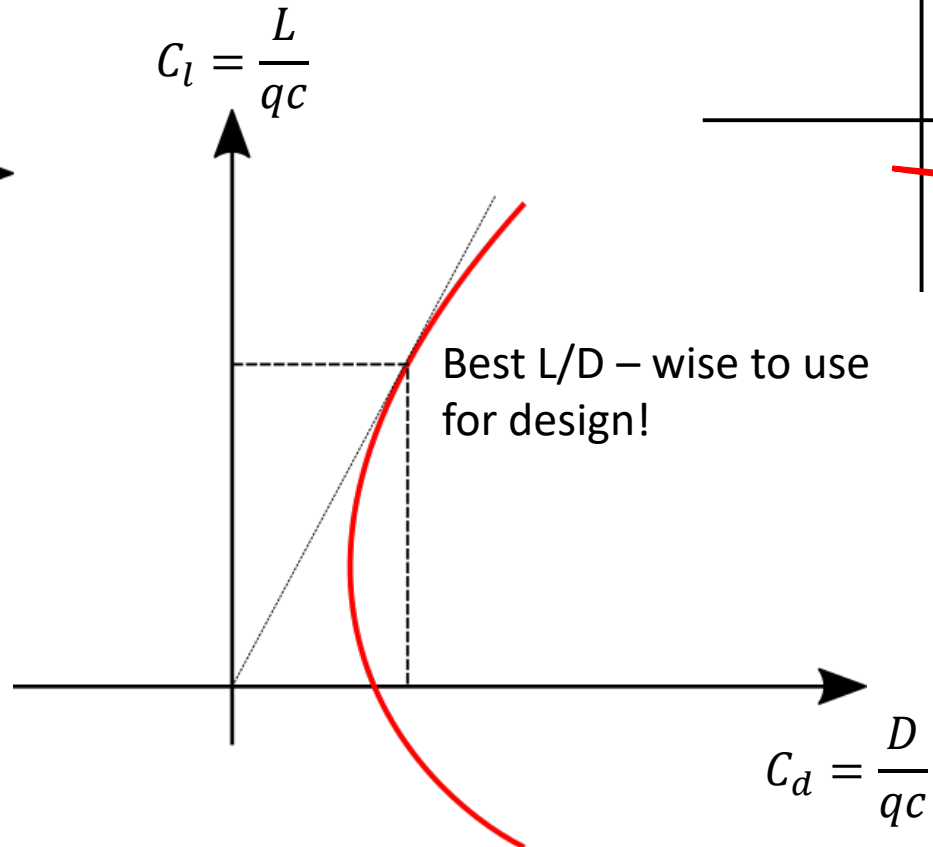
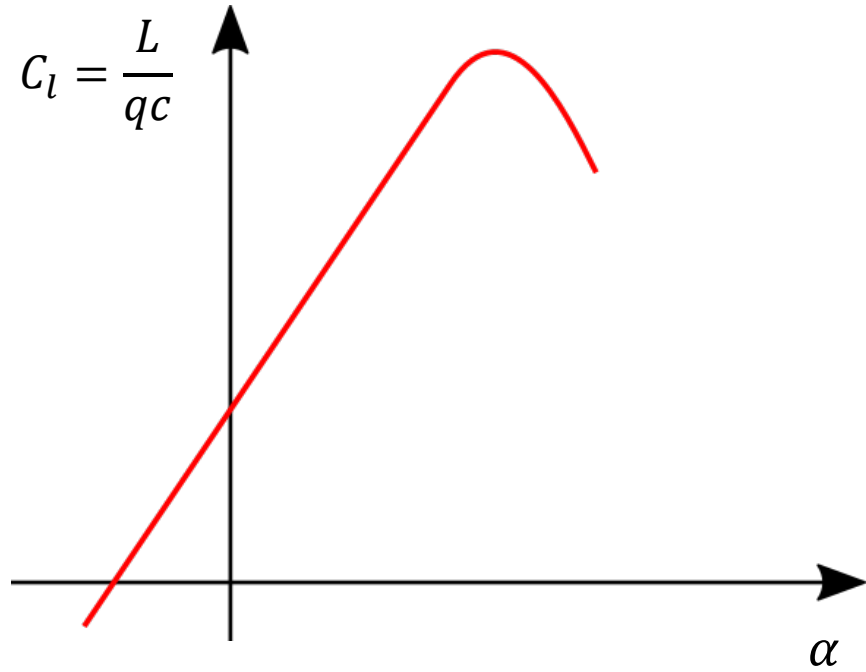
$$C_m = \frac{M'}{qc^2}$$

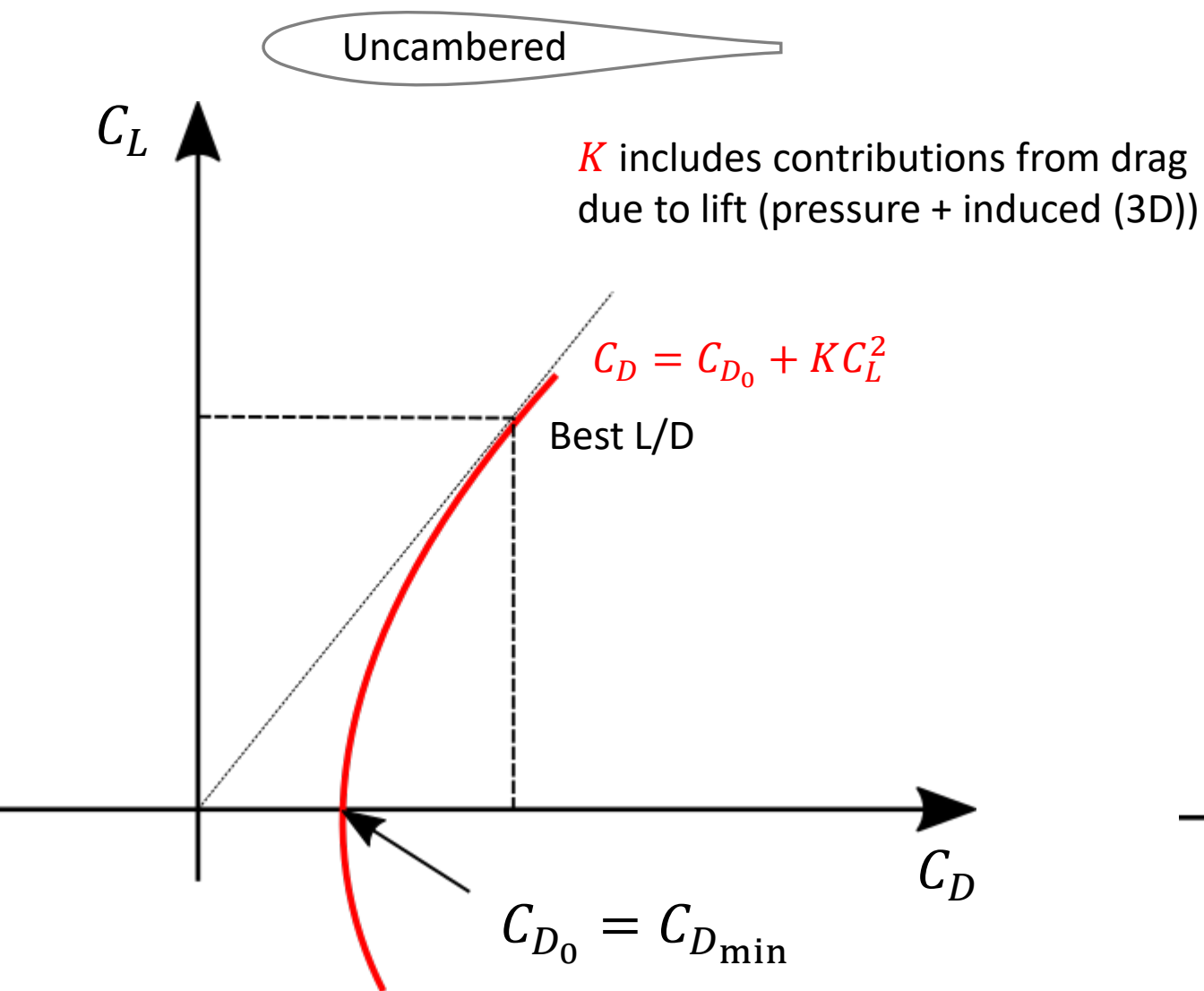


L' - Section lift (N/m)

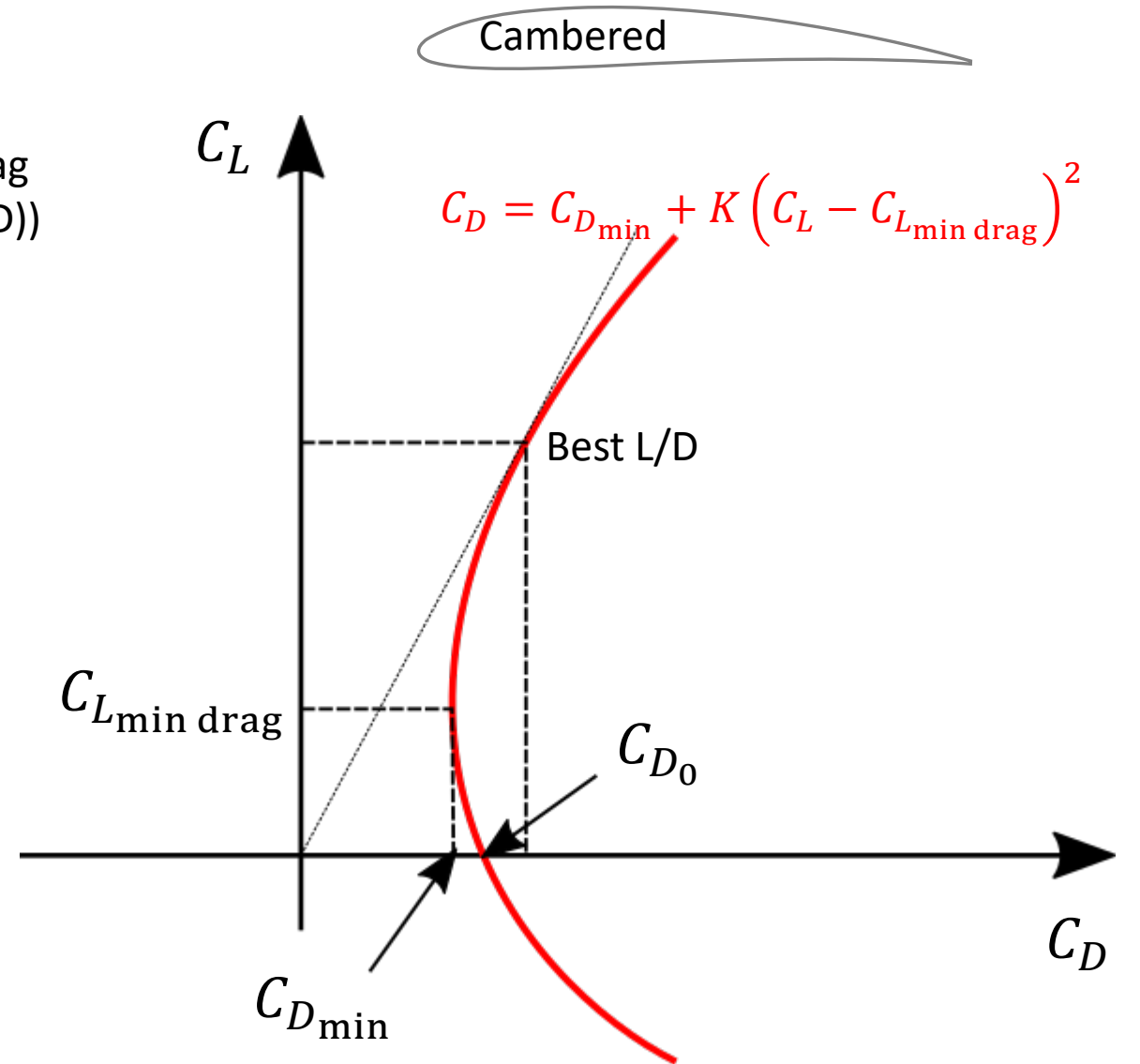
D' - Section Drag (N/m)

M' - Section pitching moment (N.m/m)





C_{D_0} - Drag coefficient for zero lift



For moderate camber $C_{D_{\min}} = C_{D_0}$

- ✈ Adequate design
 - Speed
 - Stall characteristics
 - Reynolds number
 - Structural performance

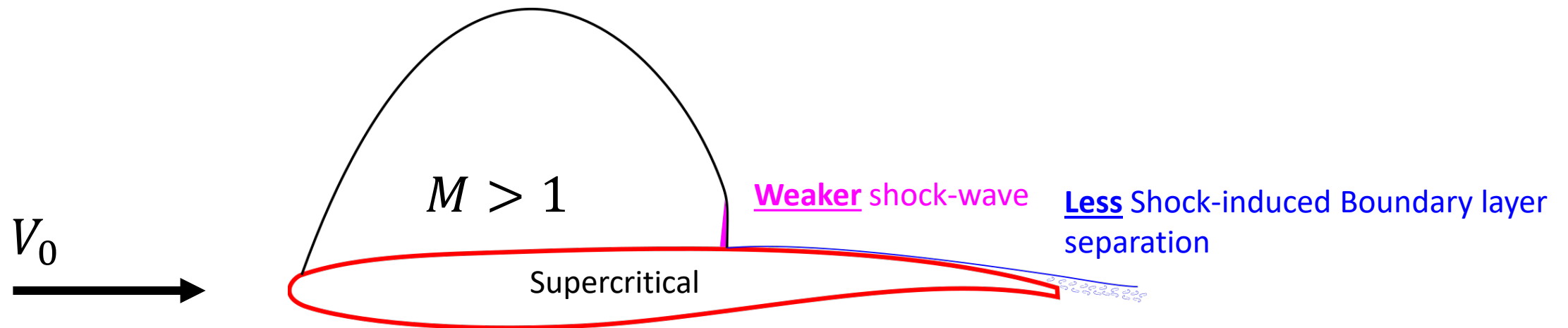
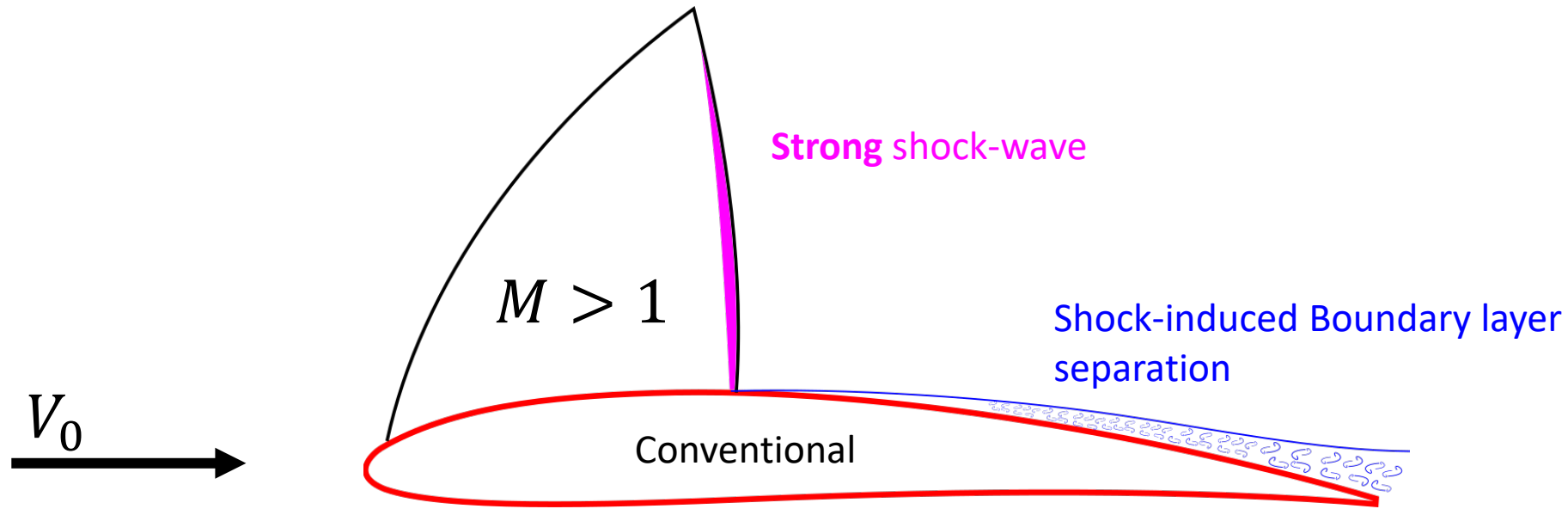
- ✈ Approximate design lift coefficient:

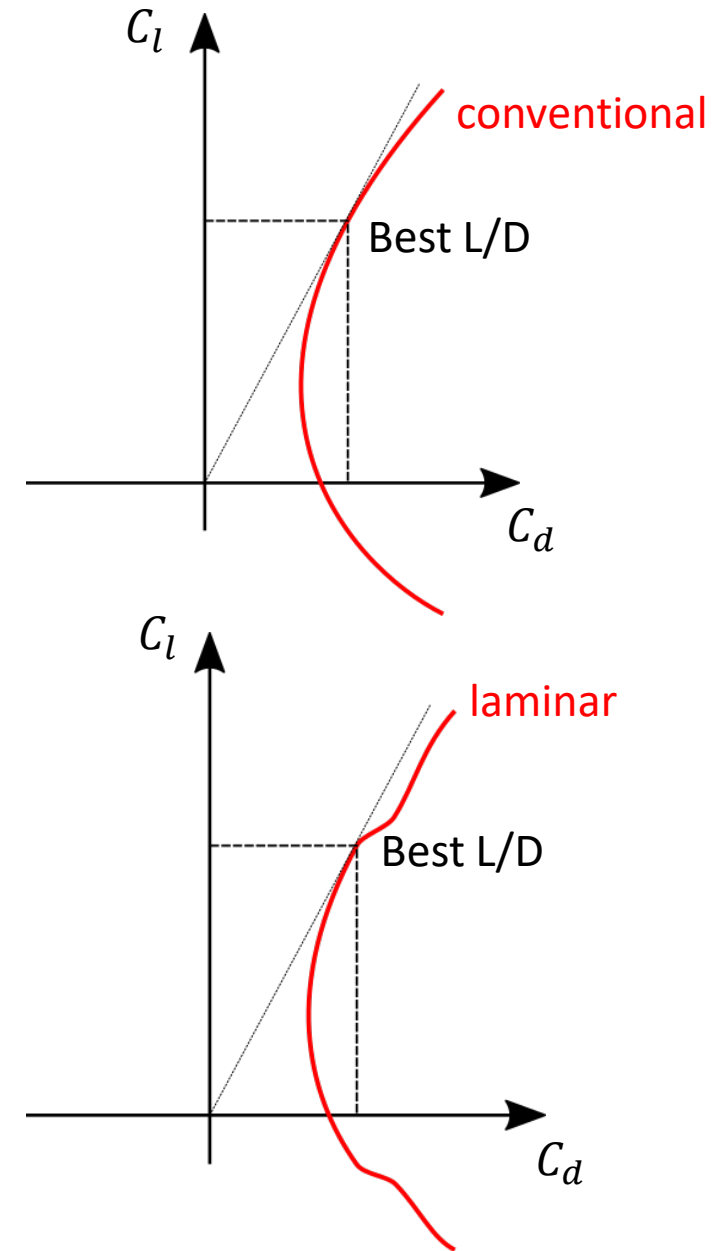
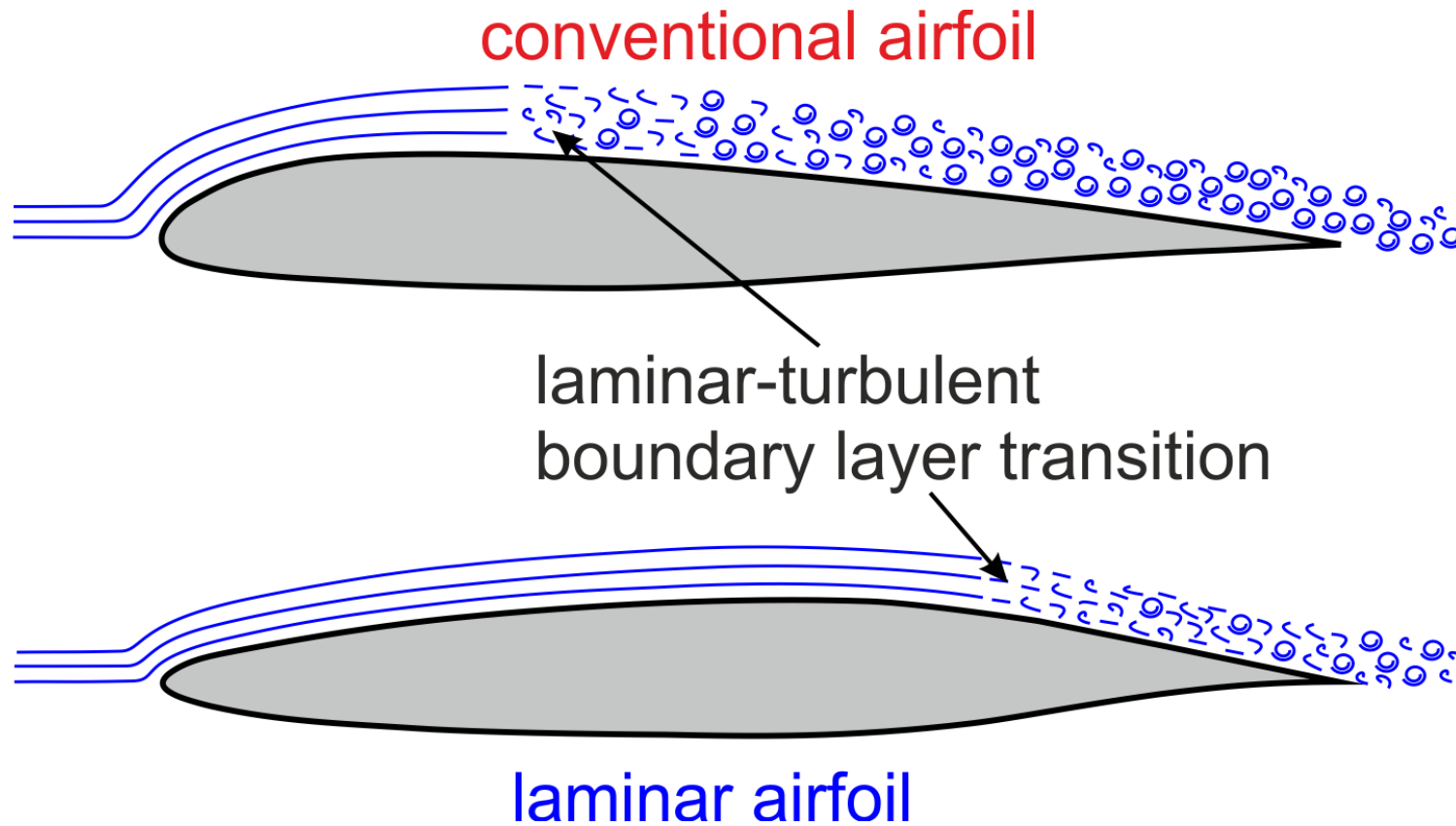
$$W = L = qSC_L \cong qSC_l$$

For a real wing 3D effects can become dominant

$$C_L = \frac{1}{q} \left(\frac{W}{S} \right)_{cruise}$$

Can be used to determine the lift coefficient at different cruise sections.
Remember the W/S reduces as you burn fuel.

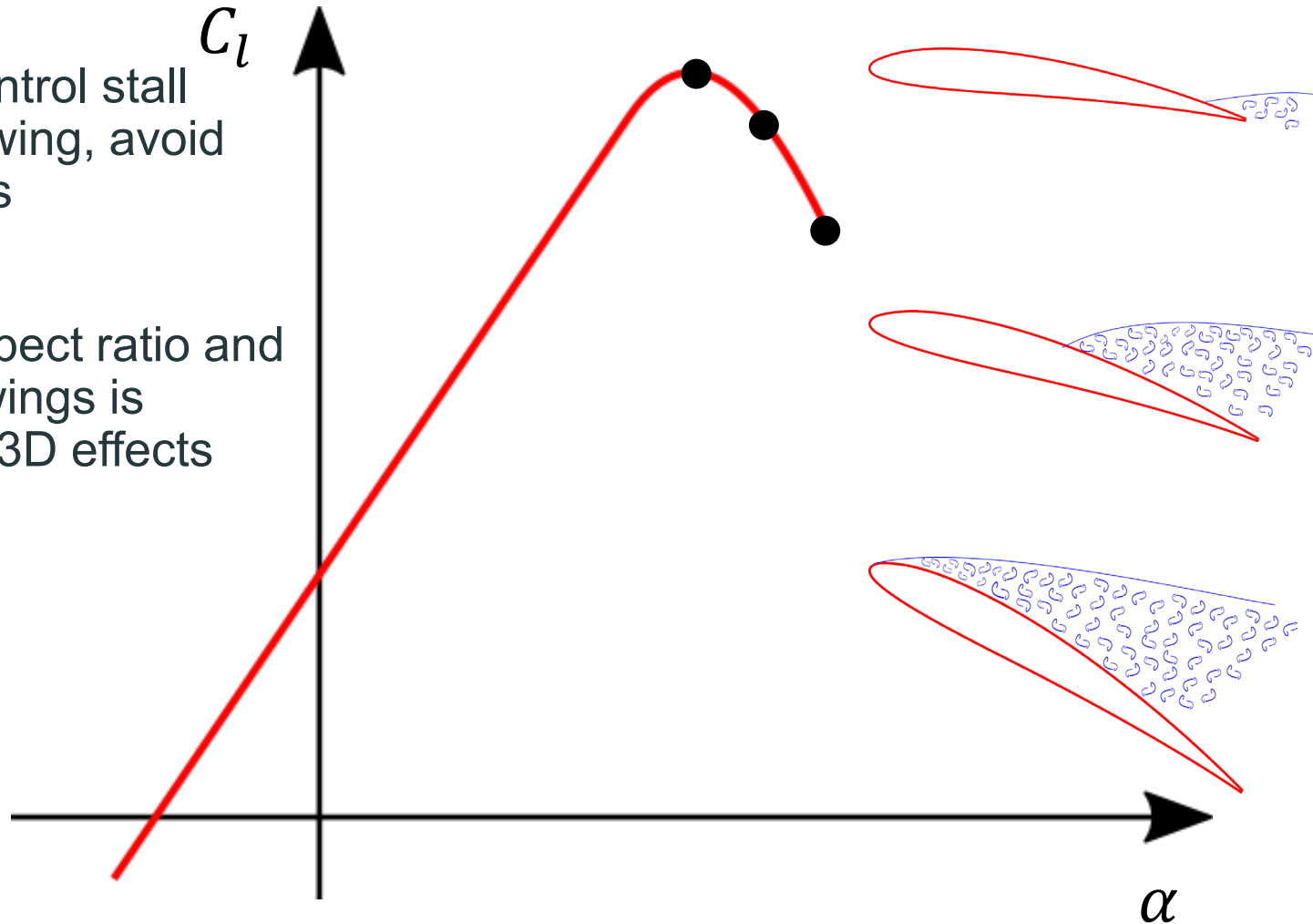




✈ Strong dependence on airfoil thickness and Re number

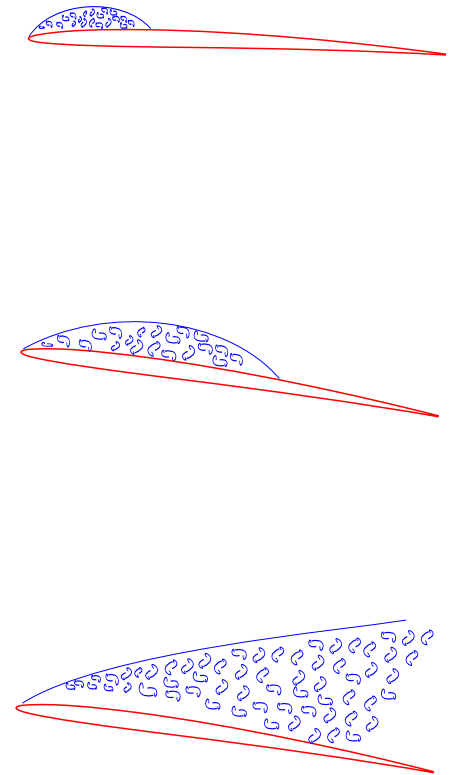
✈ Possible to control stall behavior in a wing, avoid stall in ailerons

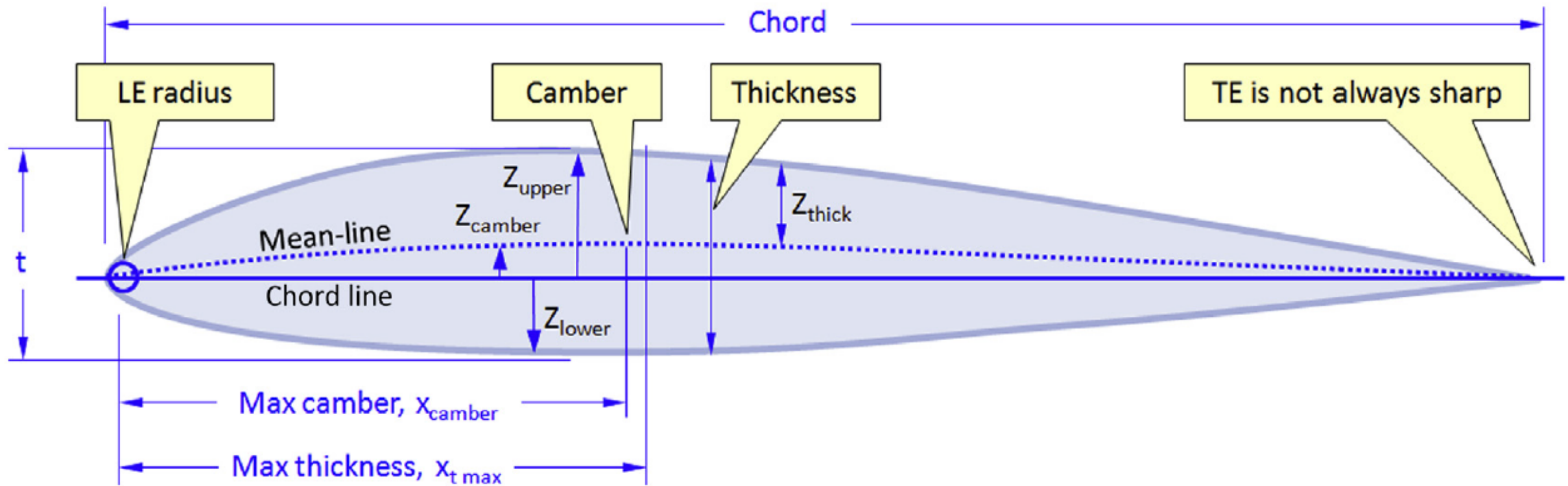
✈ Stall in low-aspect ratio and highly swept wings is dominated by 3D effects



Thick airfoil

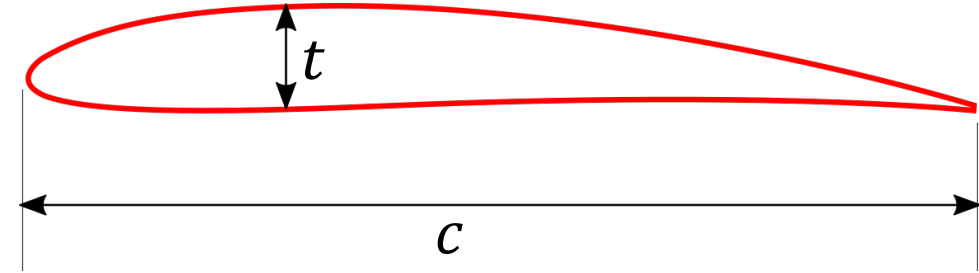
Thin airfoil





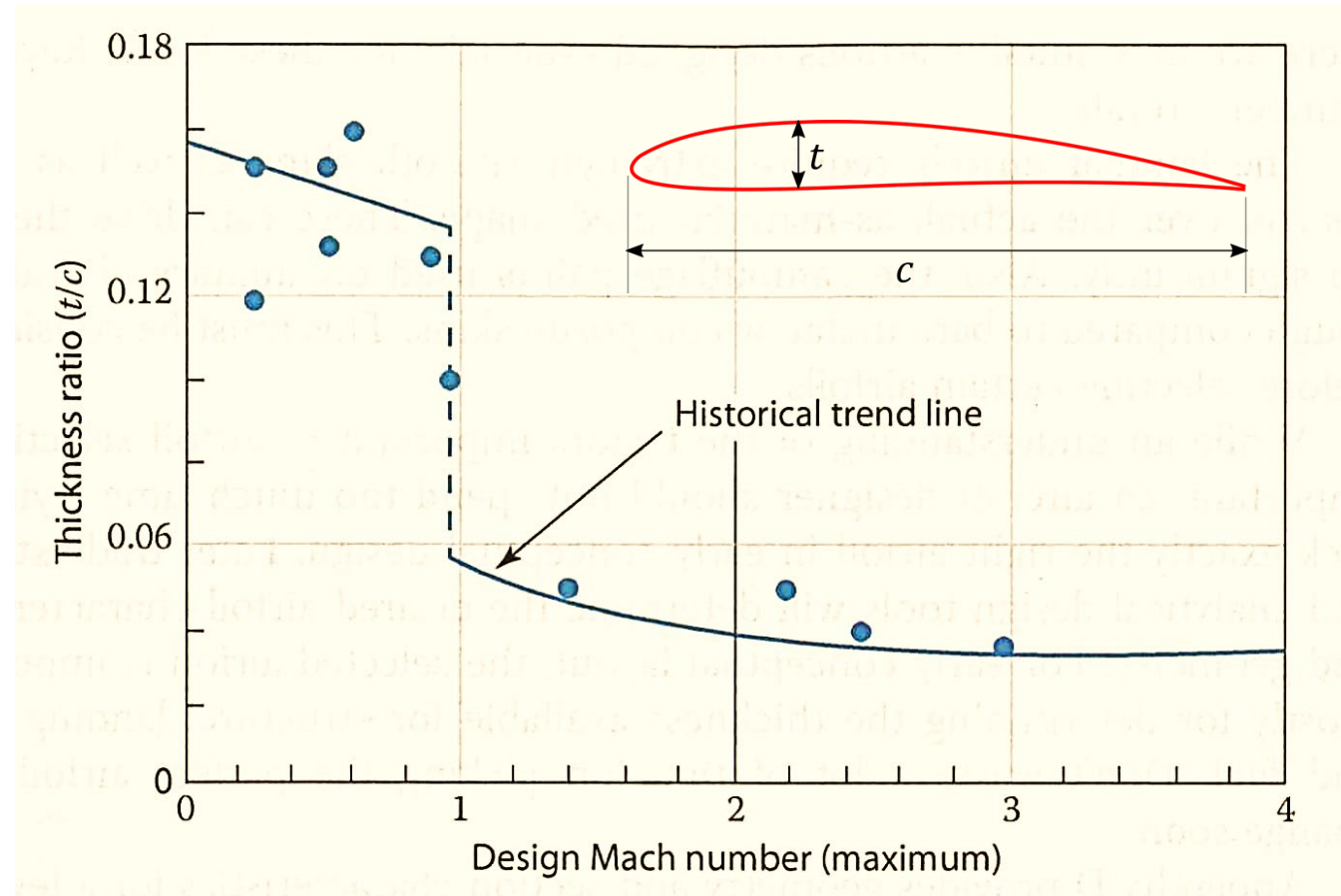
Direct impact on:

- ✈ **Drag**, larger t/c :
 - increases profile drag in subsonic flow.
 - Increases wave drag (proportional to $(t/c)^2$)
- ✈ **Maximum lift**, larger t/c :
 - Increases on high-aspect ratio wings, thicker leading edges can accommodate larger turning.
 - Decreases for low-aspect ratio, swept wings (delta wings), LE edge vortices beneficial
- ✈ **Stall characteristics**, larger t/c :
 - leads to more slowly progressing stall
- ✈ **Structural weight**, larger t/c :
 - Reduces wing mass because torsional and bending stiffness are increased.
 $\sim \frac{1}{\sqrt{t/c}}$



Initial selection:

- ✈ Use historical t/c data.
- ✈ Supercritical airfoil can be 10% thicker.
- ✈ Frequently t/c varies from root (20% - 60% larger) to tip.
- ✈ At the early conceptual phase, t/c is only important for structure, landing gear and fuel allowances.



✈ Subsonic



✈ Supersonic



- ✈ Reference wing extends to aircraft centerline.
- ✈ Manufactures have different definitions.
- ✈ Used to obtain the aerodynamic coefficients.
- ✈ Used in the wing loading calculations.
- ✈ Shape is set by taper-ratio, aspect-ratio, sweep.



Photo: Cory W. Watts



Photo: Airwolfhound



Photo: Simon_sees



Photo: Bill Abbott



Photo: Airwolfhound

✈ Taper ratio: $\lambda = c_t/c_r$

✈ Wing area: $S = 2 \int_0^{b/2} c(y) dy = b \left(\frac{c_r + c_t}{2} \right)$

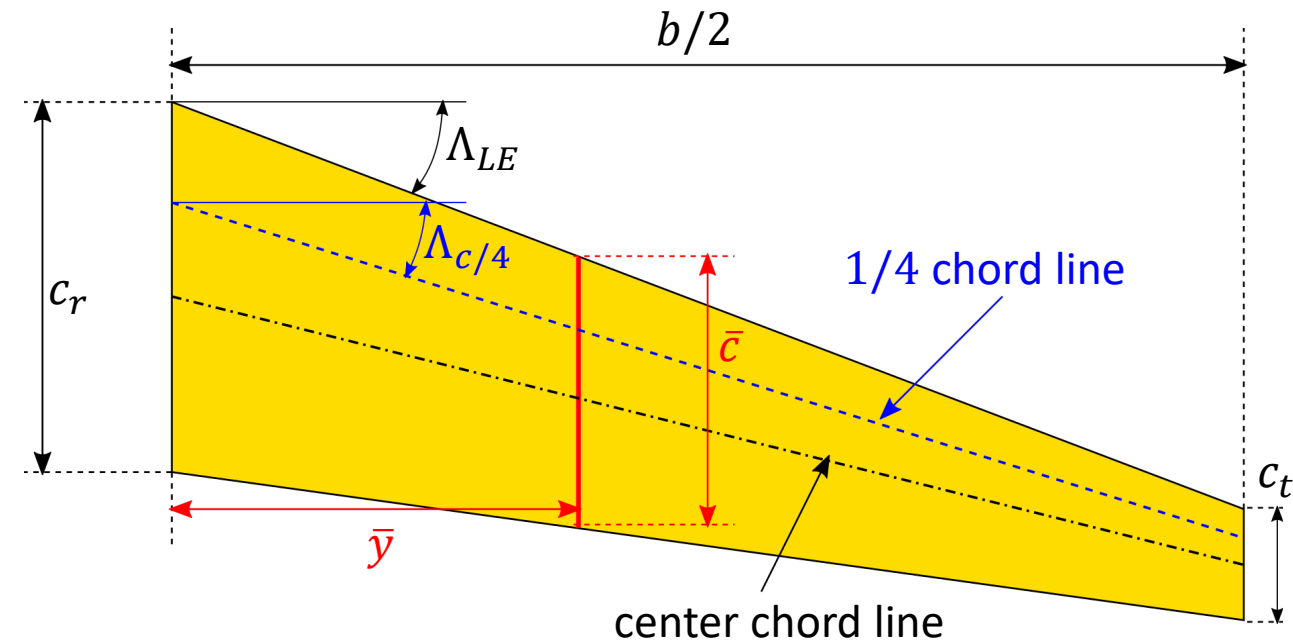
✈ Aspect ratio: $A = b^2/S$

✈ Mean aerodynamic chord (mac):

$$\bar{c} = \frac{2}{S} \int_0^{b/2} c(y) dS = \frac{2}{3} c_r \left(\frac{1 + \lambda + \lambda^2}{1 + \lambda} \right)$$

✈ Y location of mac:

$$\bar{y} = \frac{2}{S} \int_0^{b/2} y dS = \frac{b}{6} \frac{1 + 2\lambda}{1 + \lambda}$$



b – Span
 c_r – Chord at root
 c_t – Chord at tip
 $\bar{c}$ – Mean aerodynamic chord
 Λ_{LE} – Leading edge sweep angle
 $\Lambda_{c/4}$ – Quarter chord sweep angle

✦ Taper ratio: $\lambda = c_t/c_r$

✦ Wing area: $S = (c_r - c_t)y_b + (c_b + c_t)\frac{b}{2}$

✦ Aspect ratio: $A = b^2/S$

✦ Y location of mac:

$$\bar{y} = \frac{2}{S} \int_0^{b/2} y dS = \frac{2}{S} [A + B + C + D]$$

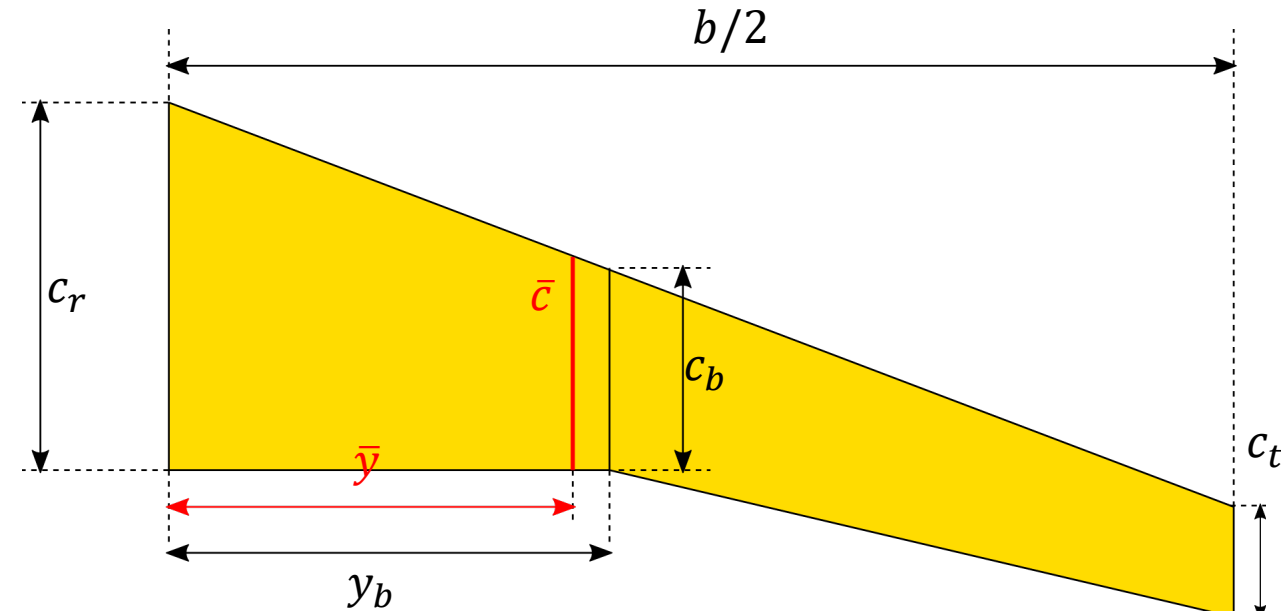
✦ Mean aerodynamic chord:

• If $\bar{y} \leq y_b$:

$$\bar{c} = c_r + (c_b - c_r) \frac{\bar{y}}{y_b}$$

• If $\bar{y} > y_b$:

$$\bar{c} = c_b + \left(\frac{c_t - c_b}{\frac{b}{2} - y_b} \right) (\bar{y} - y_b)$$



$$A = \frac{1}{2} (c_r - c_b) y_b^2$$

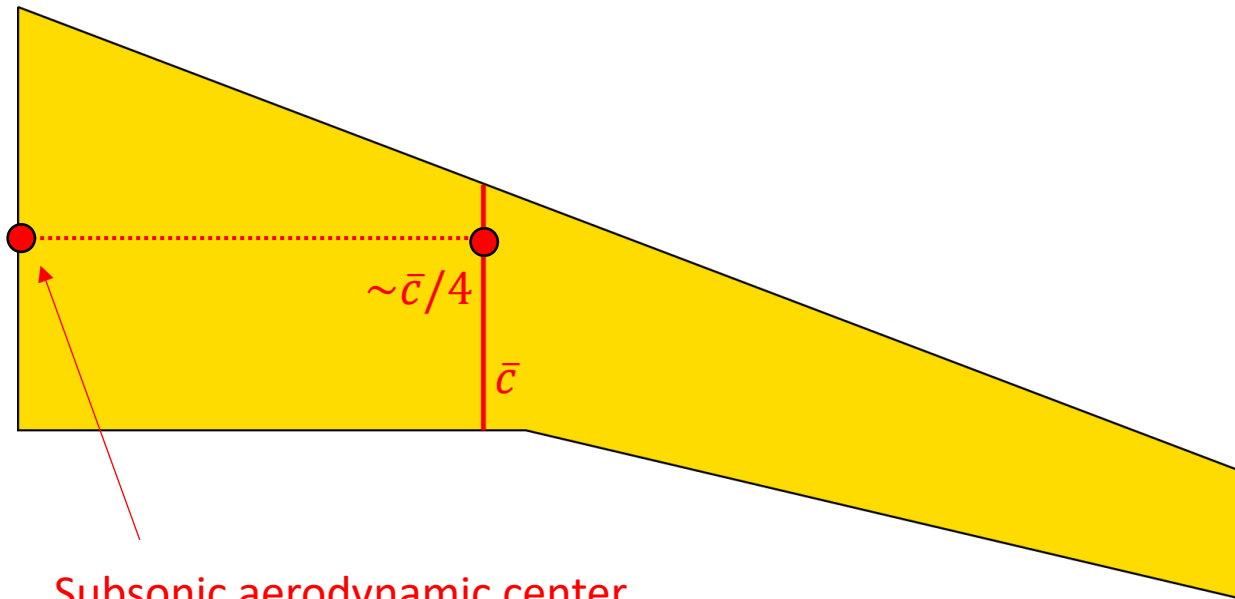
$$B = \frac{1}{3} \left[\frac{c_b - c_r}{y_b} + \frac{1}{2} a \right] y_b^3$$

$$C = \frac{1}{2} [c_b - a y_b] \left(\frac{b}{2} \right)^2$$

$$D = \frac{1}{3} a \left(\frac{b}{2} \right)^3$$

$$a = \frac{c_t - c_b}{\frac{b}{2} - y_b}$$

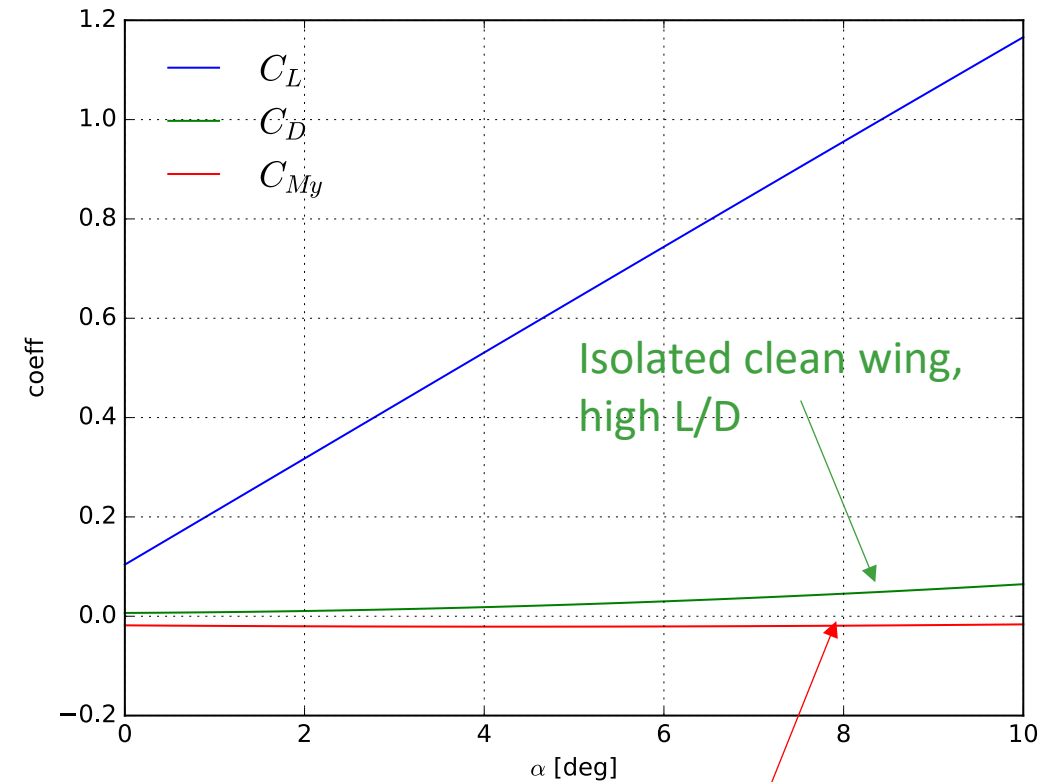
- ✈ About $\bar{c}/4$, but not exactly
- ✈ Geometry dependent
- ✈ Useful for aircraft stability



Subsonic aerodynamic center

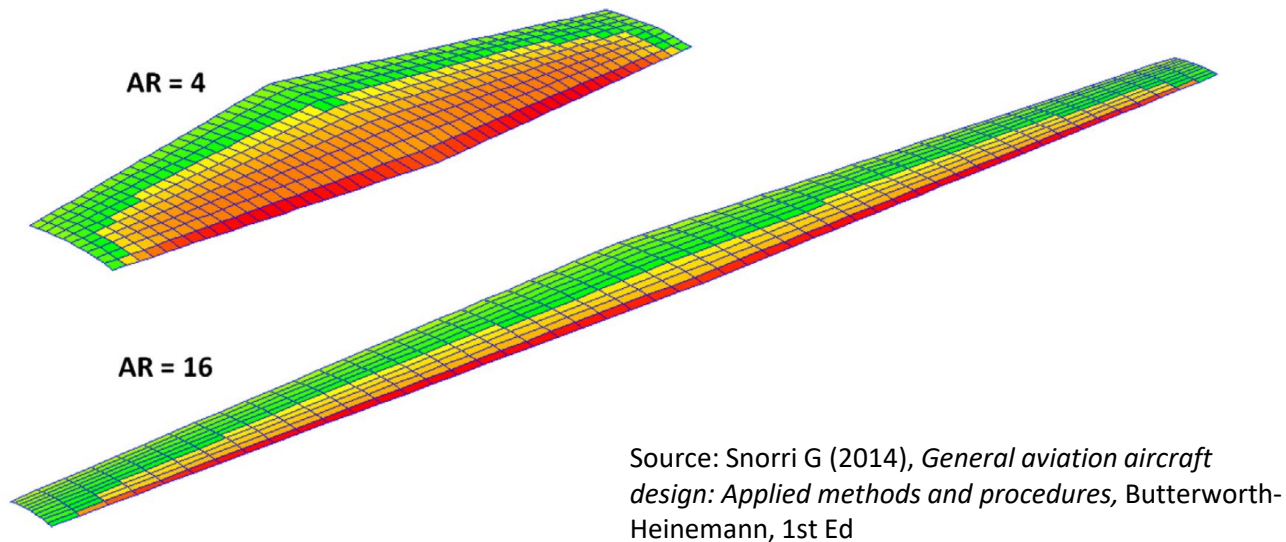
Note: Only for the wing

Calculation with OpenVSP

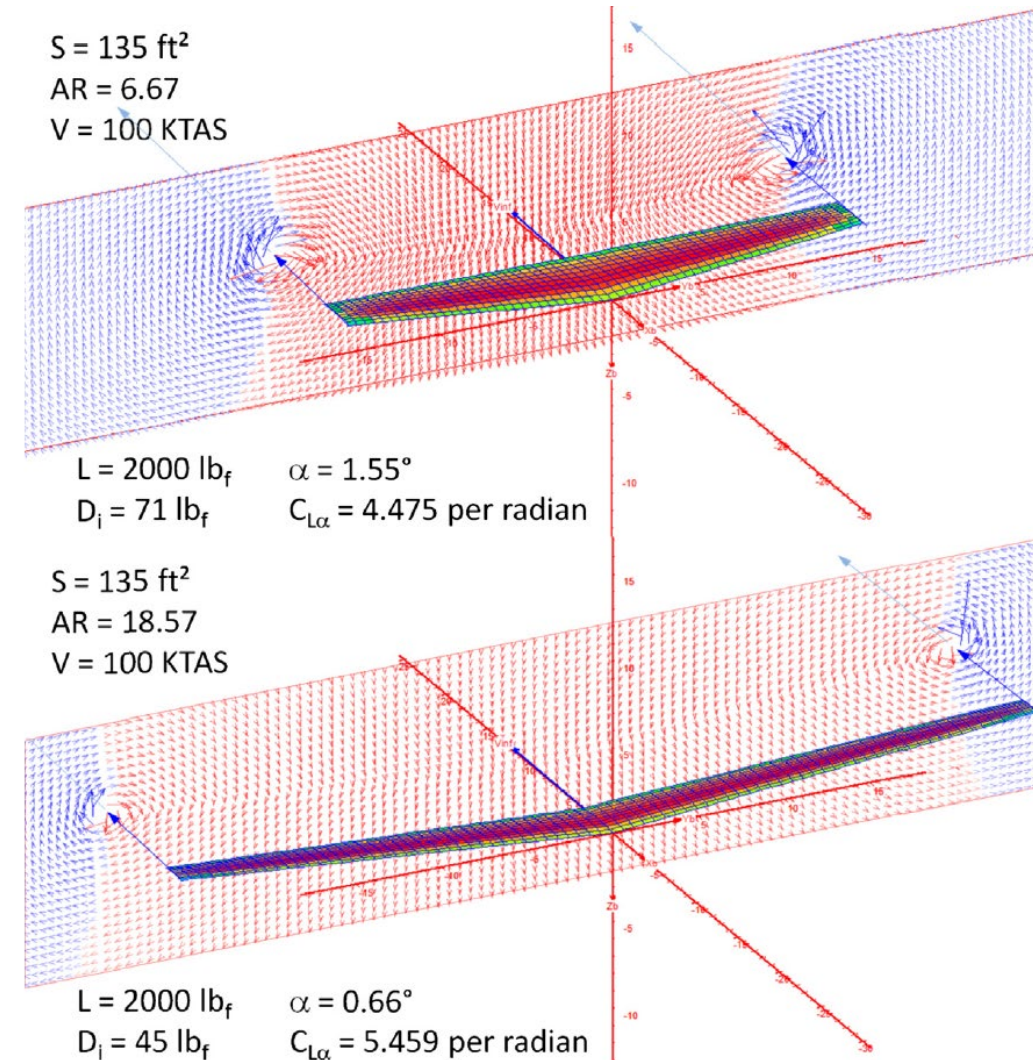


Small variation of pitch moment with AoA

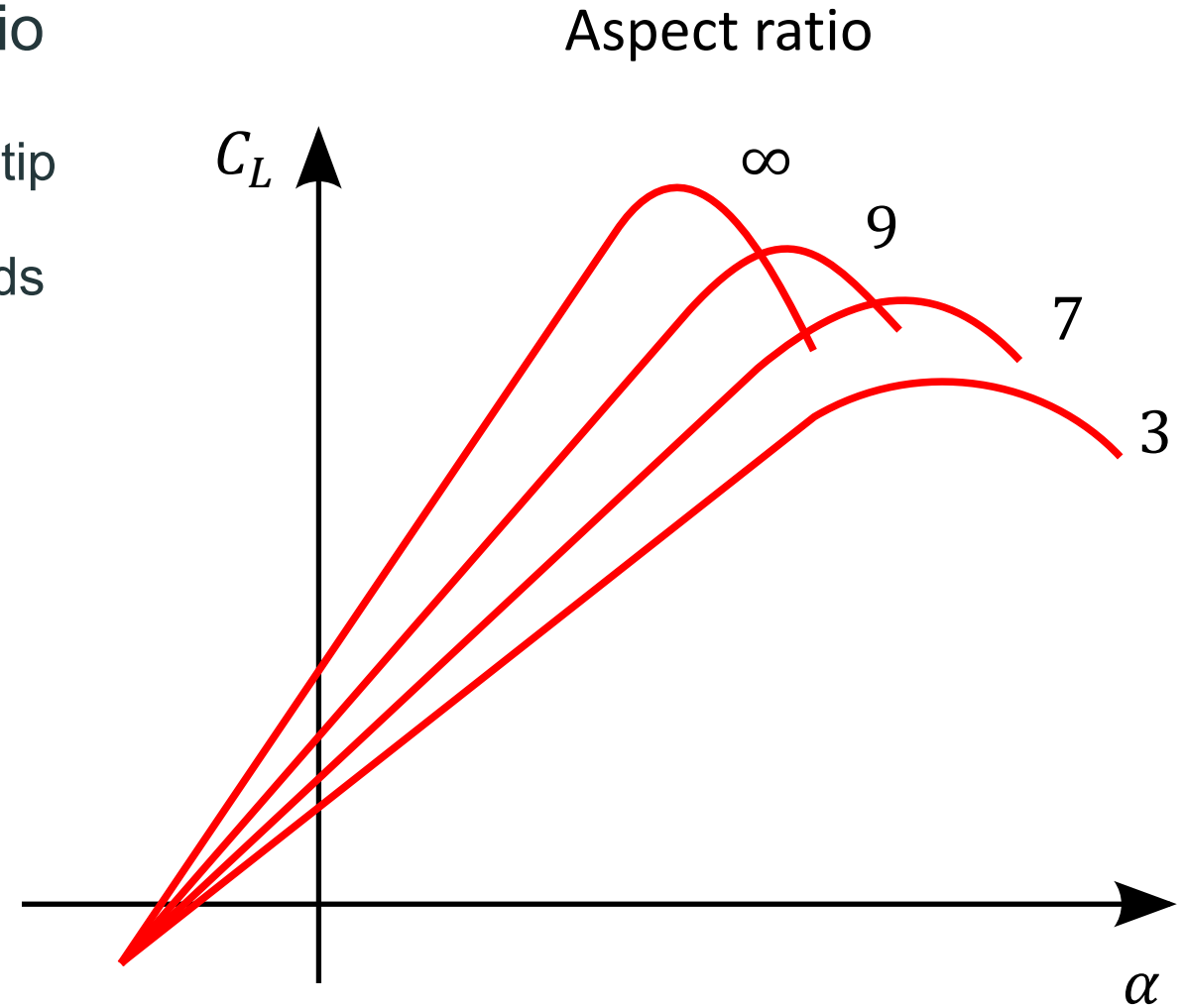
- ✈ For the same area, increasing aspect-ratio
 - Improves L/D by reducing induced drag.



Planforms of equal area and different aspect ratio



- ✈ For the same area, increasing aspect-ratio
- Improves L/D by reducing induced drag.
 - Increases maximum lift (relative lift drop in the tip is lower)
 - Increases wing weight, structural stiffness needs to be added
 - Decreases stall angle (relative changes in α at the tips)



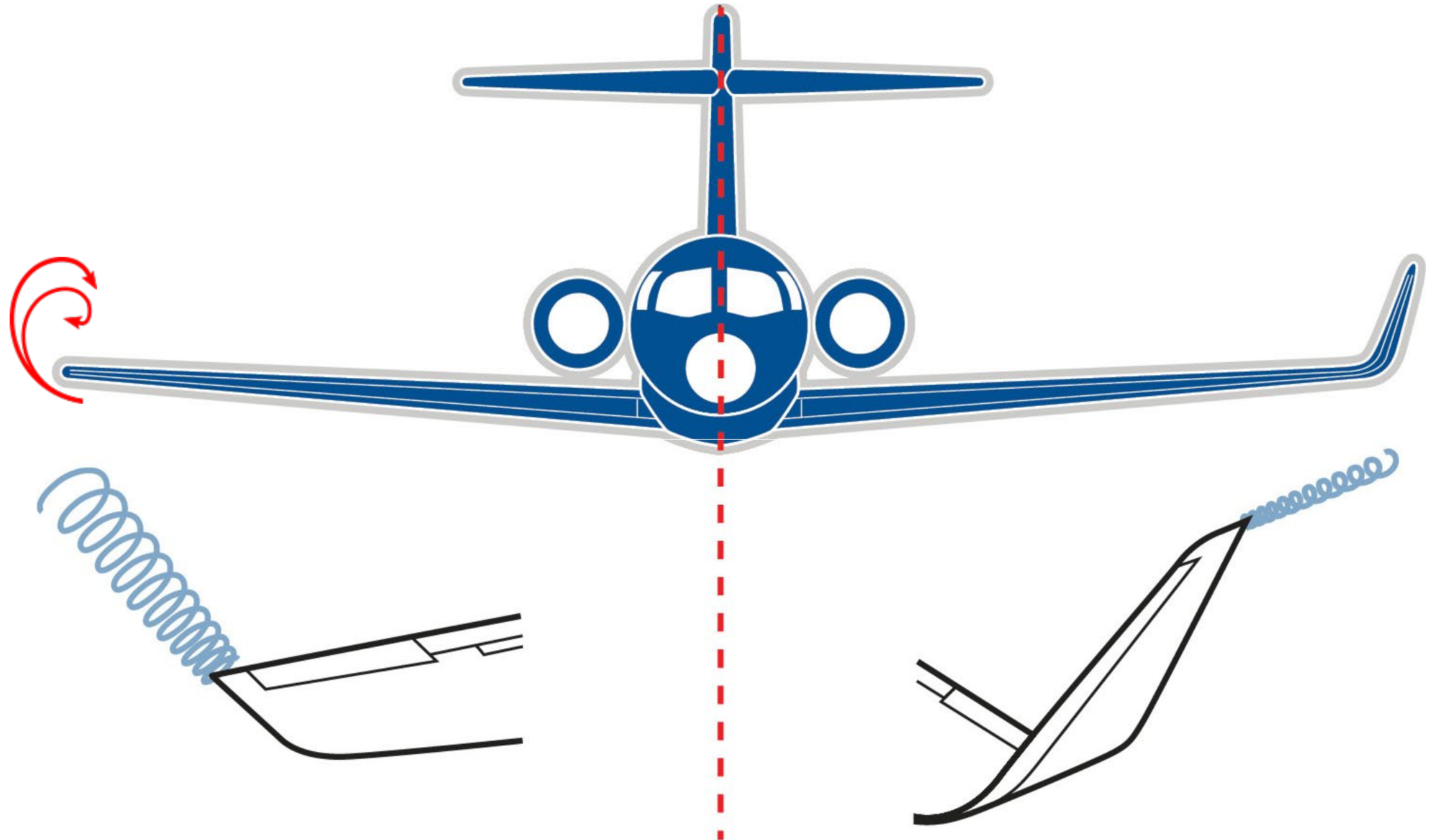
✈ For initial selection use:

$aM_{\max}^c$		
	a	c
Jet fighter	4.110	-0.622
Military Cargo	5.570	-1.075
Jet Transport	7.5 - 10	0

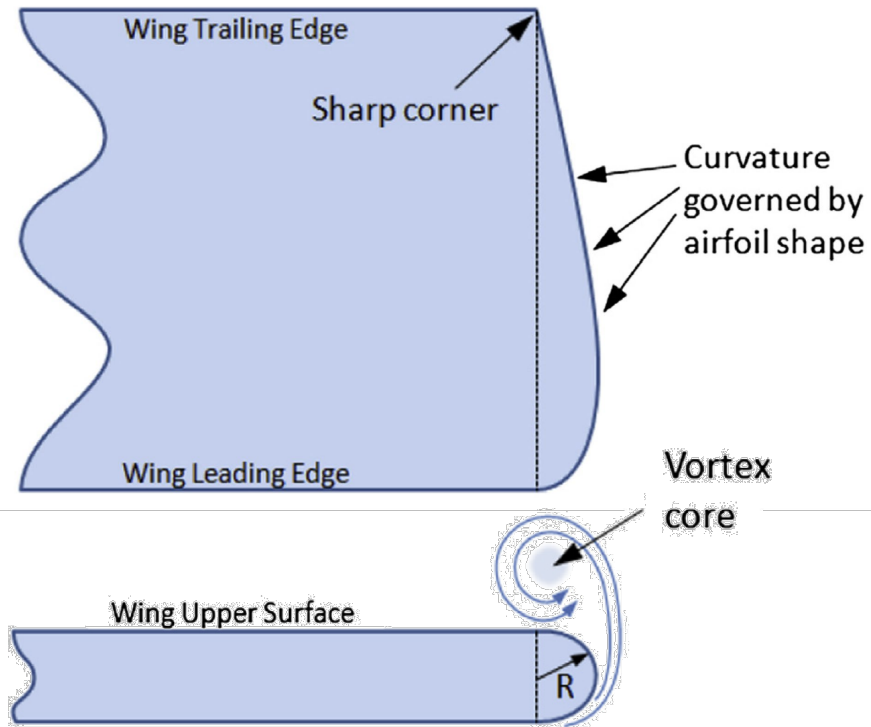
✈ Based on equivalent wing area: wing + lifting canard (if any)

✈ If designing a turboprop look into existing design (Q400, $A = 12.5$)

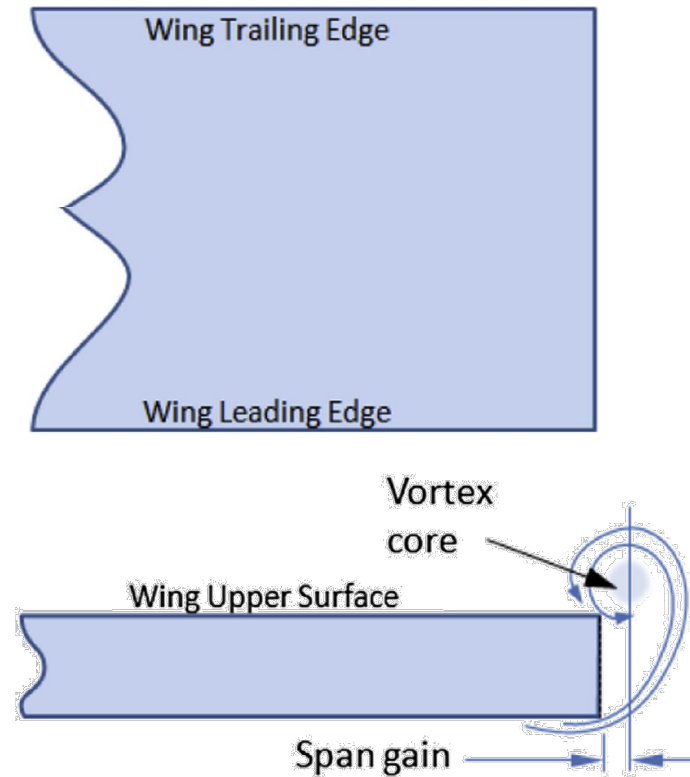
✈ Later a trade study should be done to determine the best aerodynamic vs structural weight design!



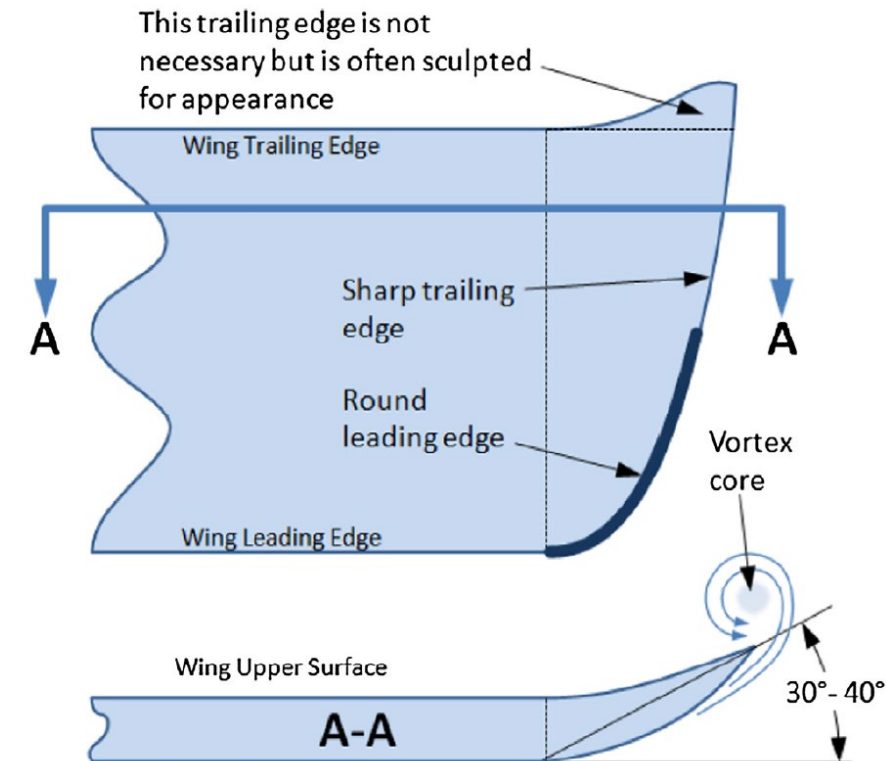
Rounded



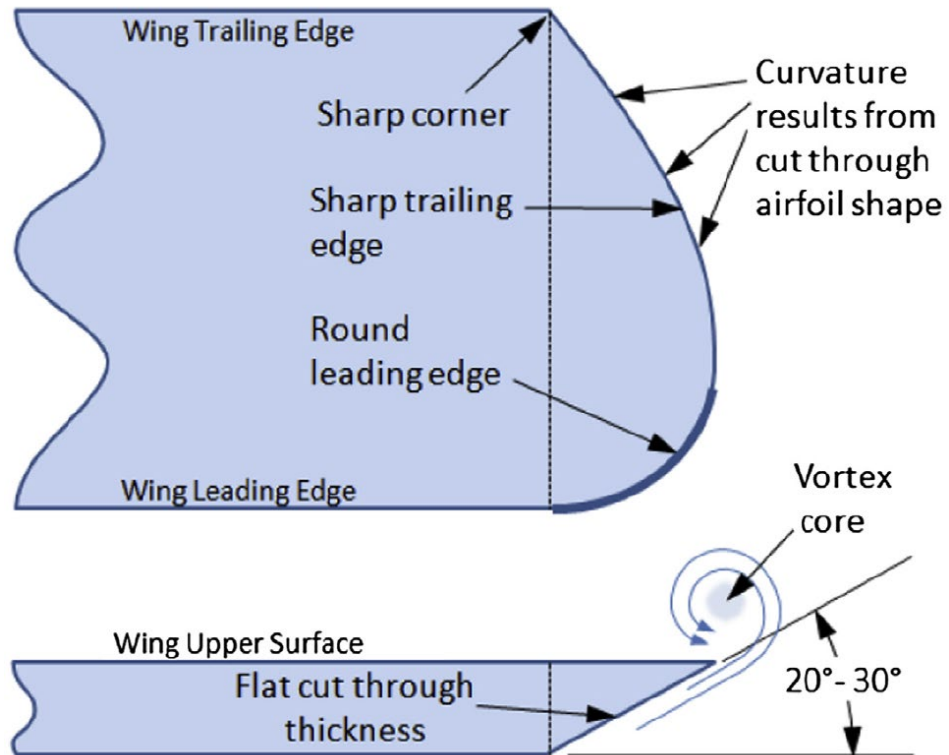
Squared



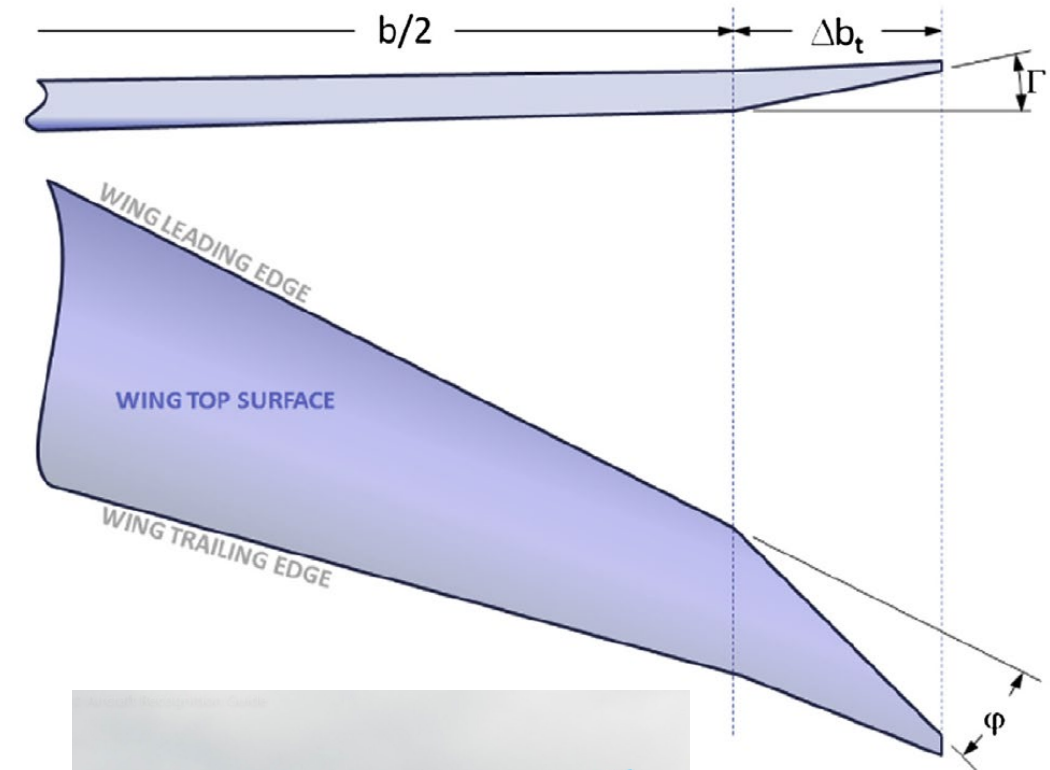
Upswept



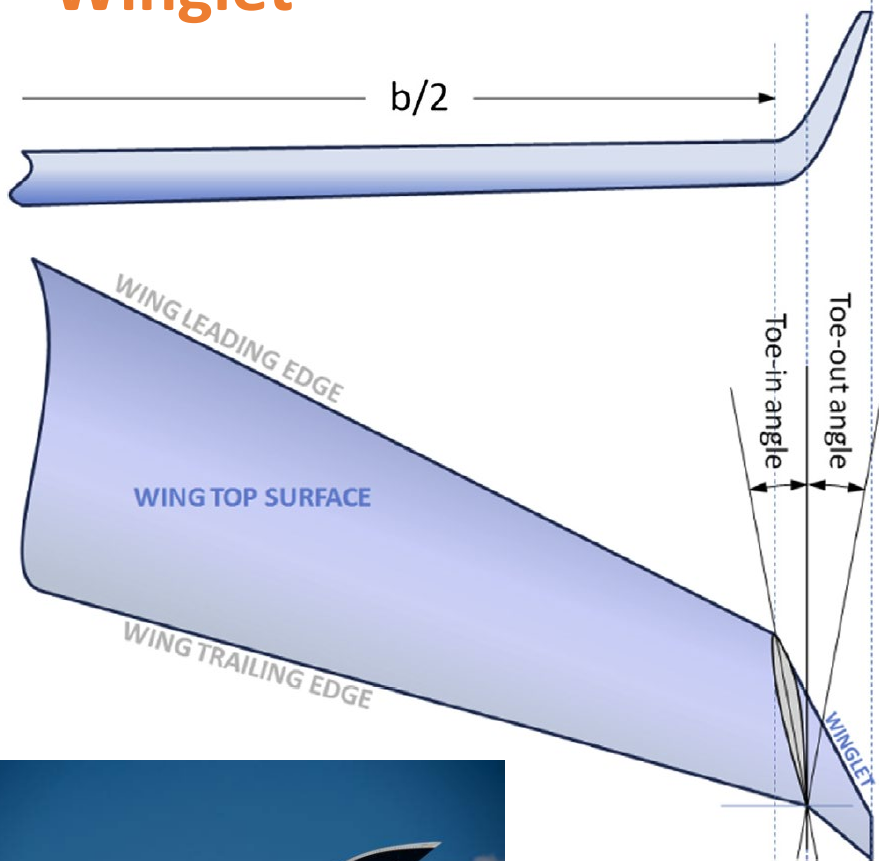
Hoerner



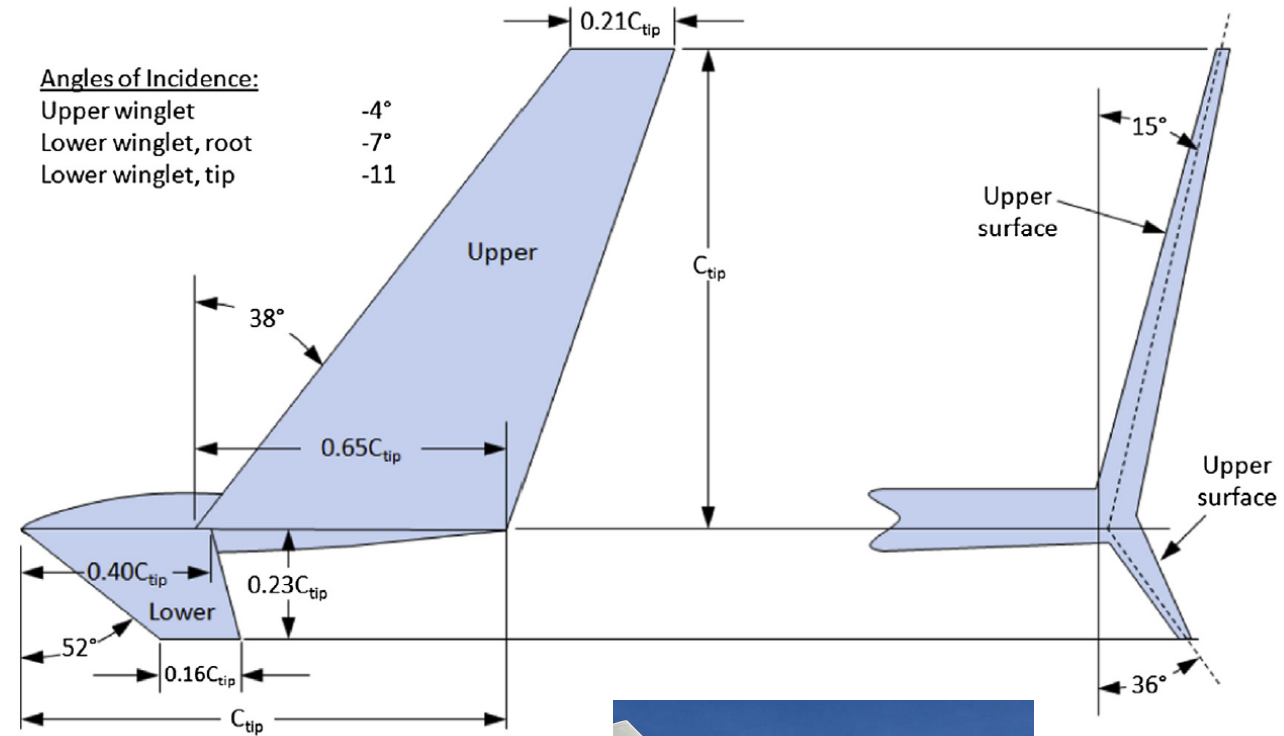
Raked



Winglet



Whitcomb Winglet



- ✈ Important for transonic and supersonic aircraft.
 - Shock formation dictated by component normal to leading edge.

$$V_N = \cos(\Lambda) V$$

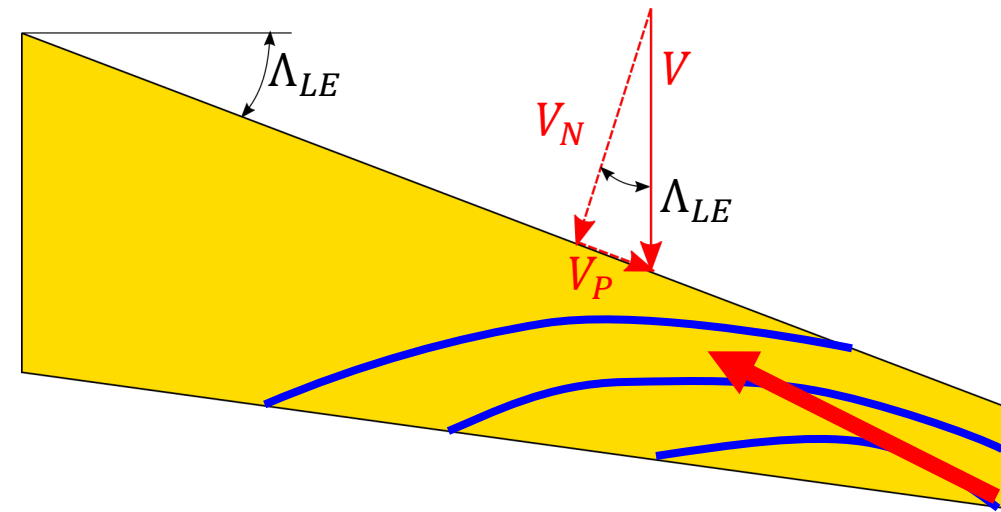
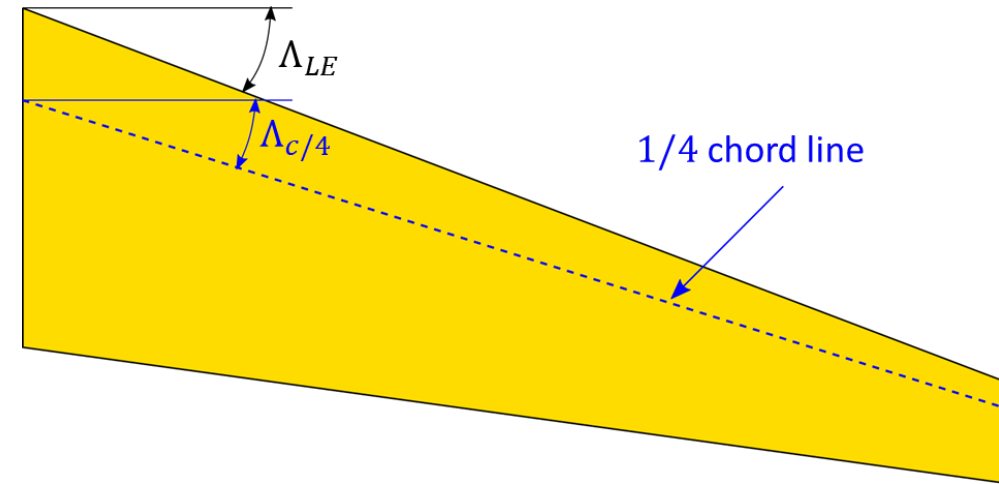
Hence:

$$M_{crit} = \frac{(M_{crit})_{\Lambda=0}}{\cos(\Lambda)}$$

M_{crit} – Far field speed at which $M > 1$ over the wing

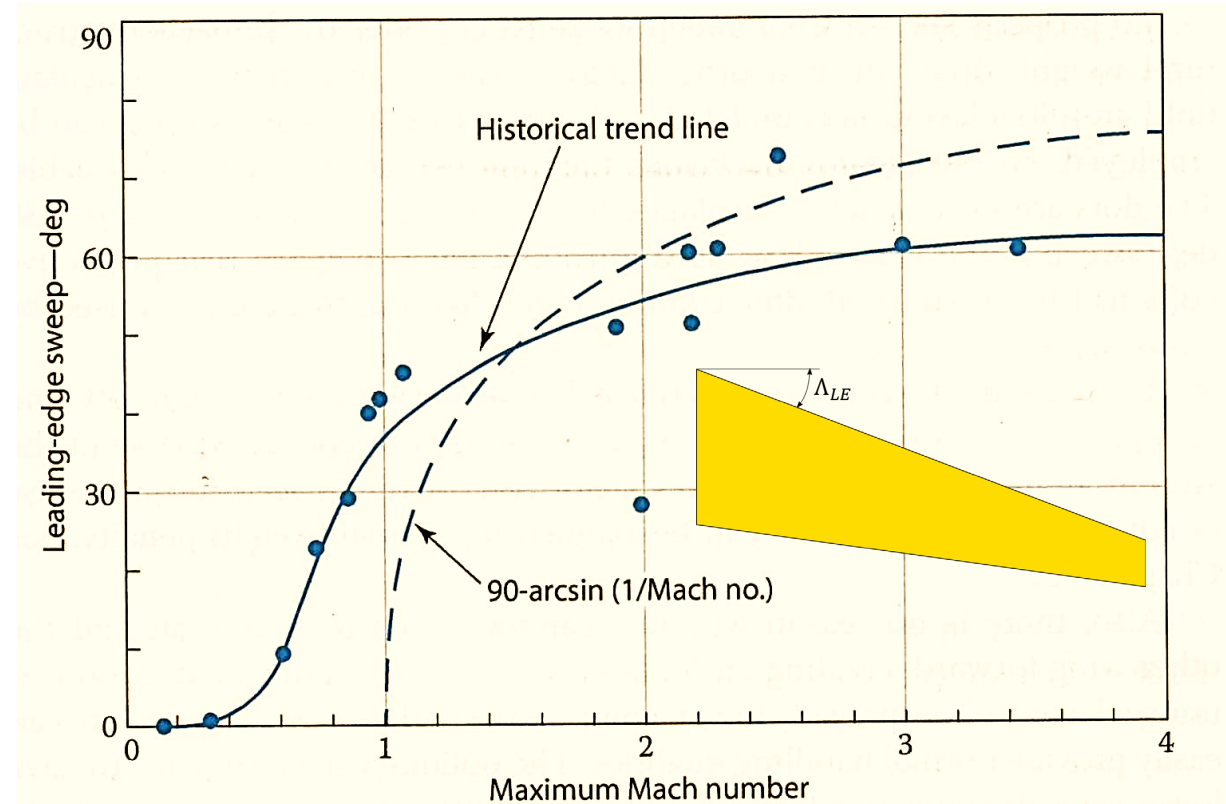
Increasing sweep:

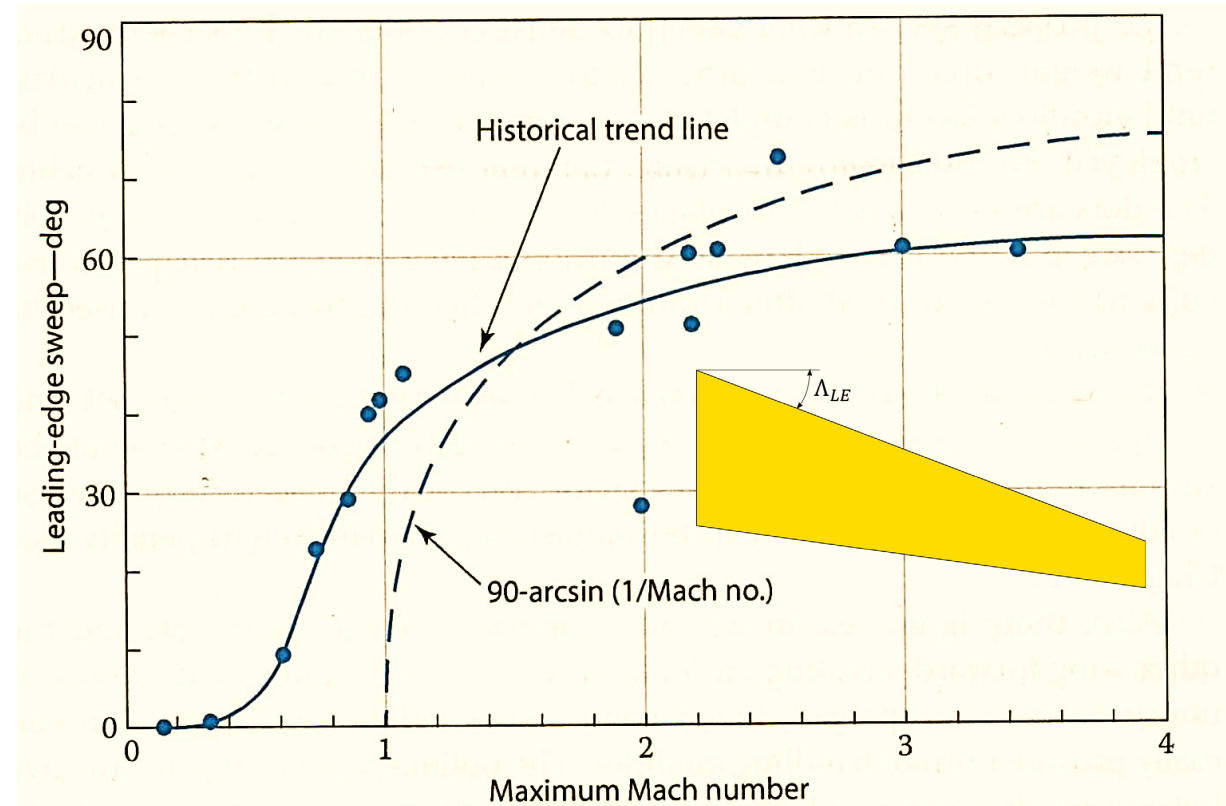
- ✈ Decreases C_{Lmax}
- ✈ Increases structural weight, center of lift moves aft
- ✈ Stall is now governed by 3D effects



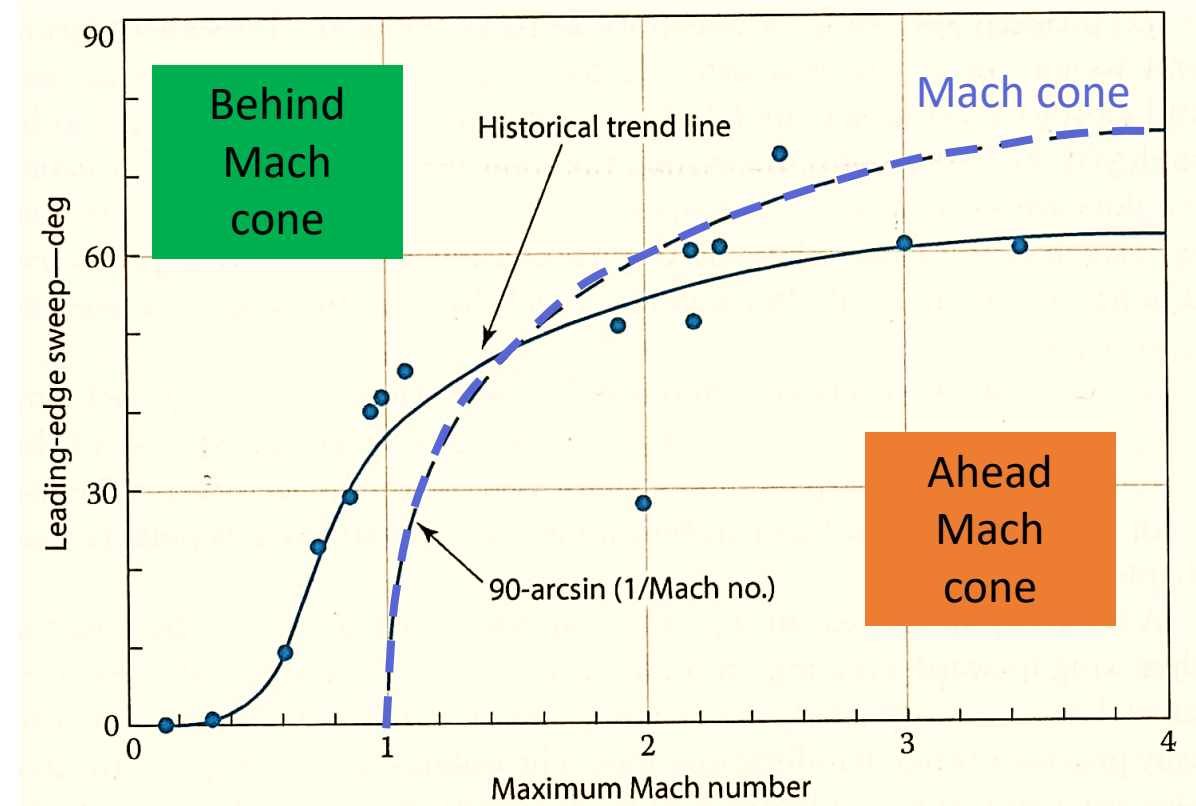
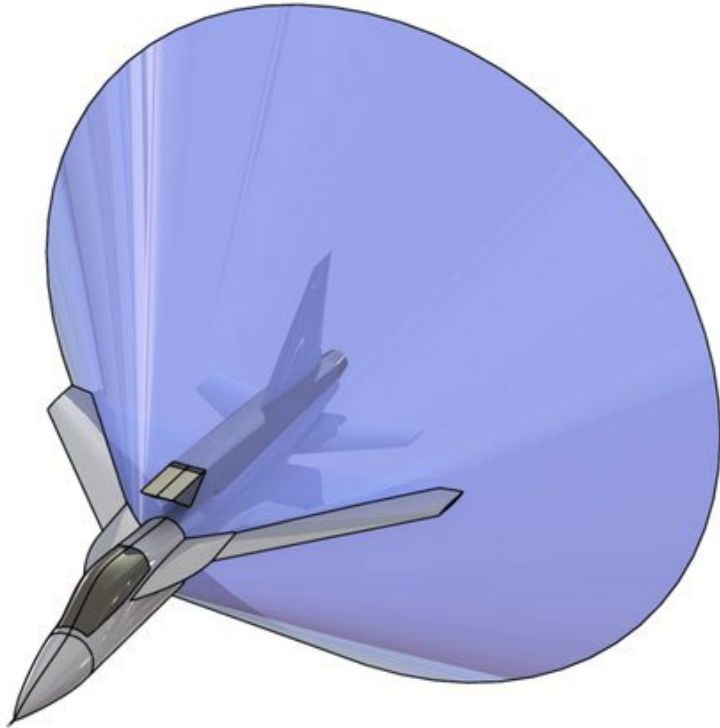
Initial selection:

- ✈ Use historical Λ data.
- ✈ For subsonic Λ is selected for the M_{crit} or M_{dd} (drag divergence Mach).
- ✈ Ultimately will be a combination of structural and aerodynamic performance.





Mach cone



Wing geometry – **taper-ratio** $\lambda = (c_t/c_r)$

- ✈ Important for lift distribution.
 - Reduce amount of load at the tip.
- ✈ Elliptical is the optimum shape (minimum drag due to lift).
- ✈ Elliptical is expensive, apply a taper ratio gives a good approximation.
- ✈ Sweep-aft requires a reduction of taper-ratio
- ✈ First approximation in Fig. 4.24 Raymer



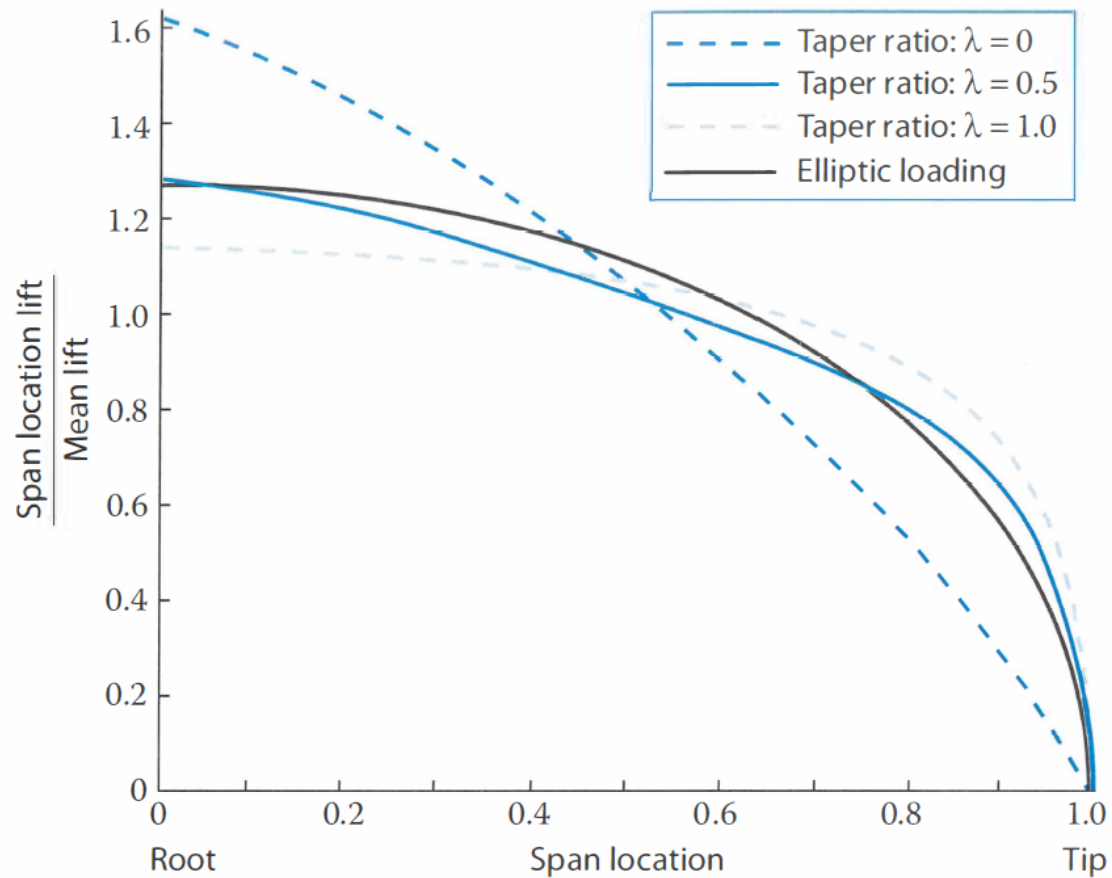


Fig. 4.23 Effect of taper on lift distribution.

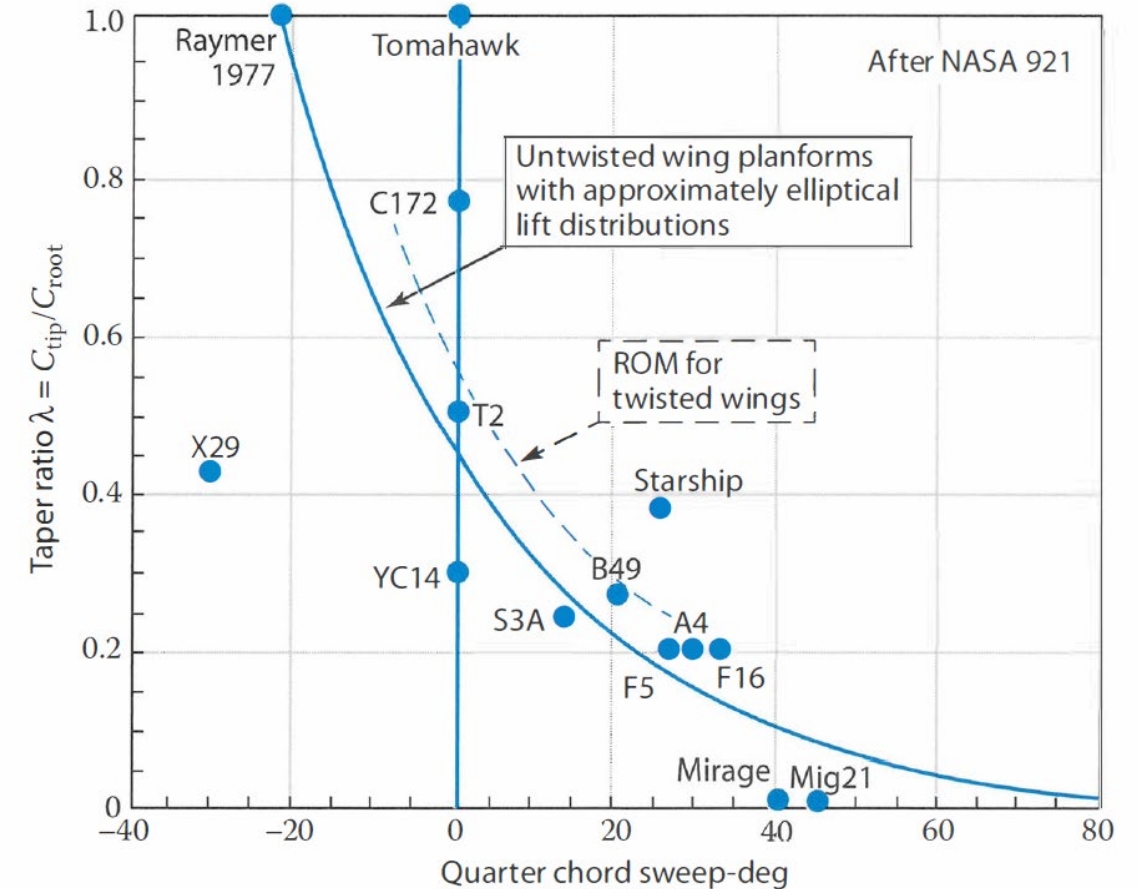
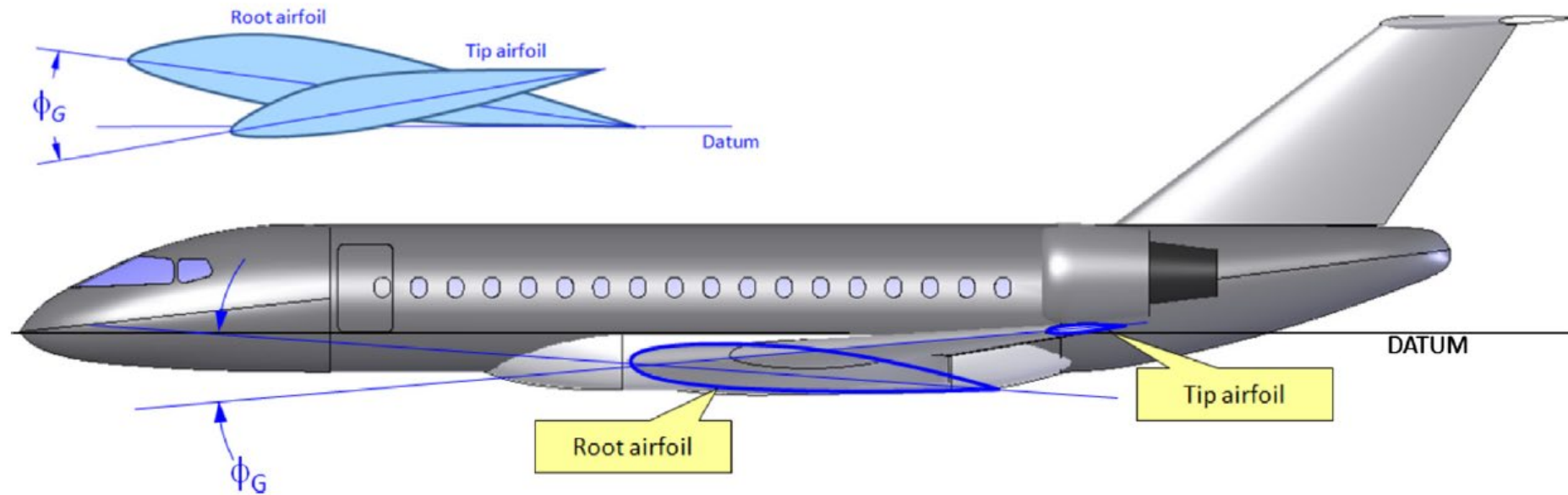
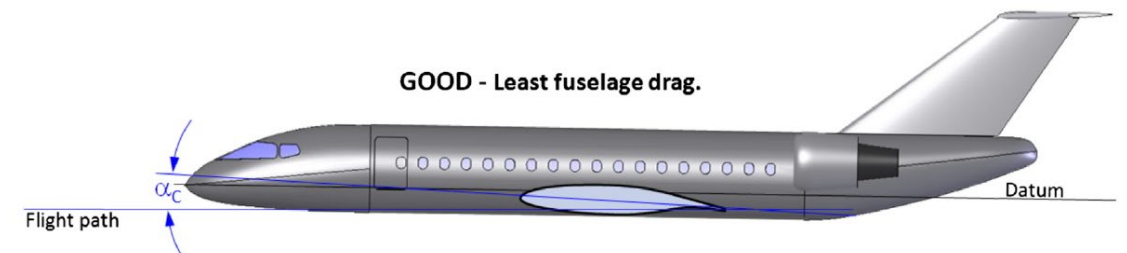
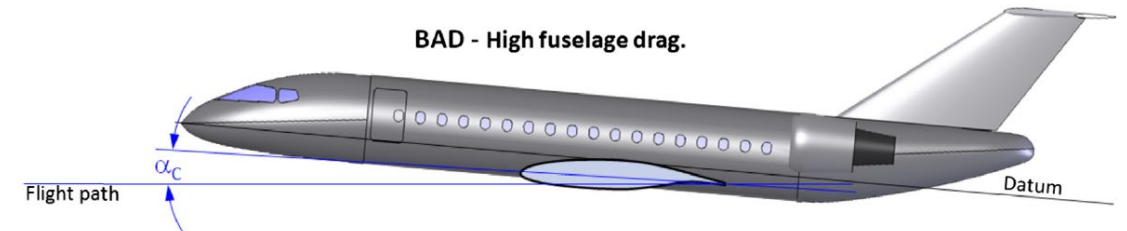
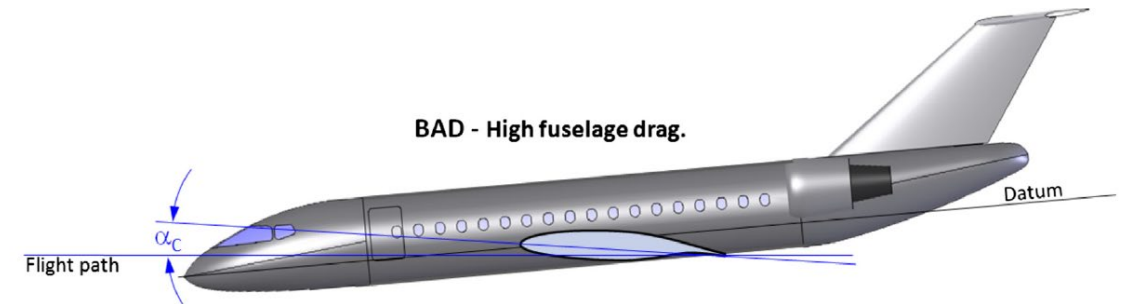
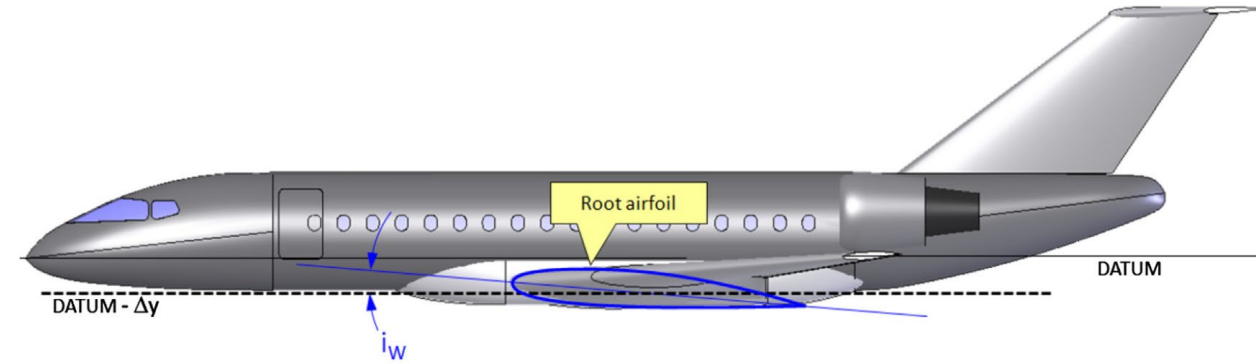


Fig. 4.24 Effect of sweep on desired taper ratio.

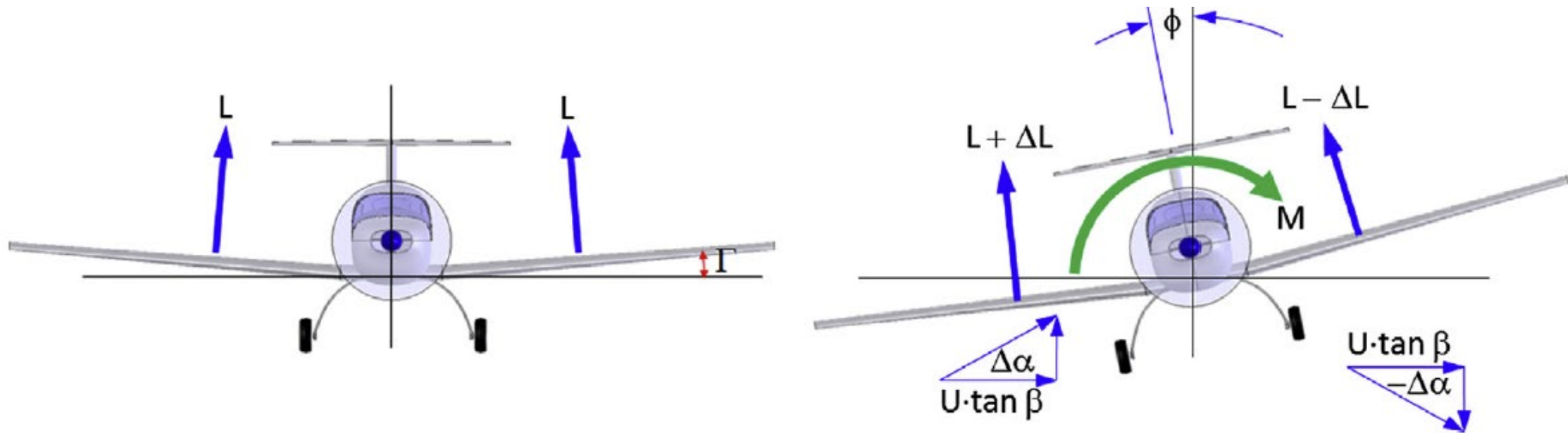


- ✈ Useful to prevent tip stall, typically between 0 and -5 deg.
- ✈ Geometric twist = Actual change in angle of incidence.
- ✈ Aerodynamic twist = Change between zero lift angles.
- ✈ Hard to establish optimal. First guess could be -3 deg (but do research)

- ✈ Pitch angle of the wing with respect to the fuselage.
- ✈ For an untwisted wing = angle between fuselage and airfoil chord-line
- ✈ Used to minimize fuselage drag at cruise.
- ✈ Set by CFD or wind tunnel data.
- ✈ Assume 1 deg for transport aircraft; 2 deg for general aviation. (but do research)

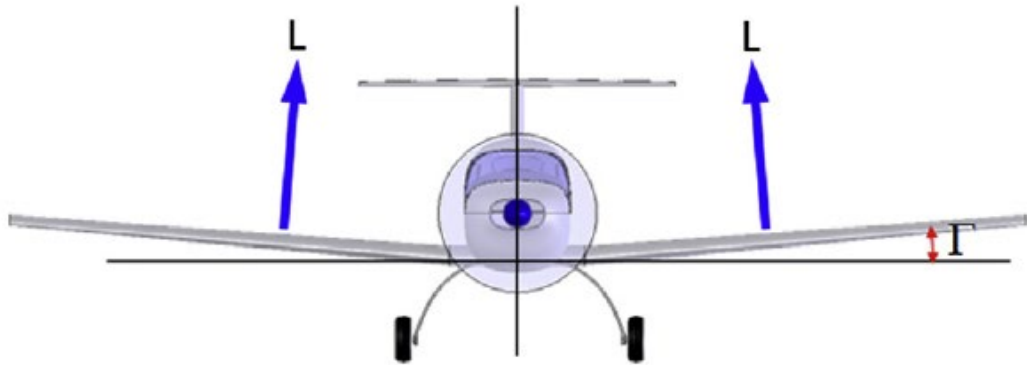


- ✈ Major contributor to roll stability
- ✈ Varies with vertical location and sweep angle
- ✈ Guidelines in table 4.2, Raymer



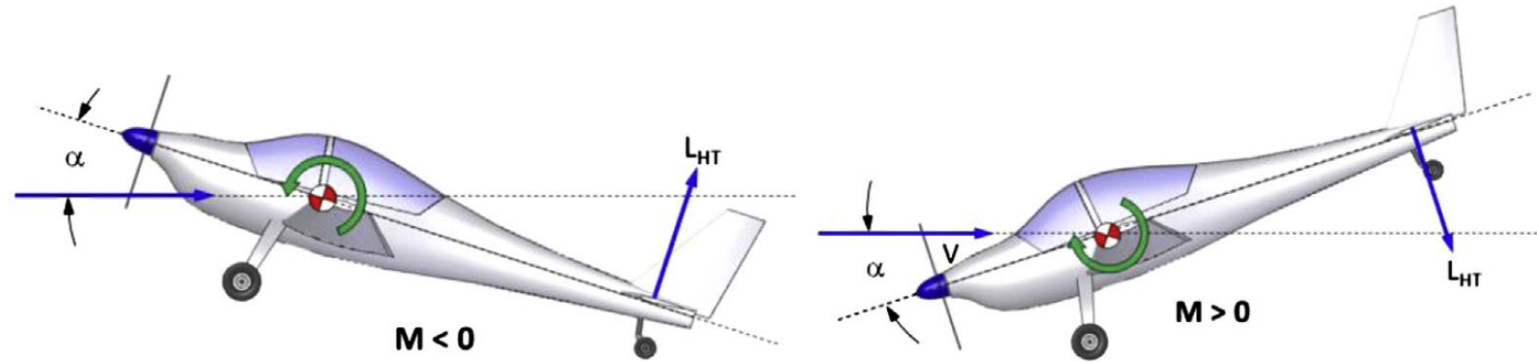
Source: Snorri G (2014), *General aviation aircraft design: Applied methods and procedures*, Butterworth-Heinemann, 1st Ed.

- ✈ Major contributor to roll stability
- ✈ Varies with vertical location and sweep angle
- ✈ Guidelines in table 4.2, Raymer



	Wing position		
	Low	Mid	High
Unswapt (deg)	5 to 7	2 to 4	0 to 2
Subsonic swept (deg)	3 to 7	-2 to 2	-5 to -2
Supersonic swept (deg)	0 to 5	-5 to 0	-5 to 0

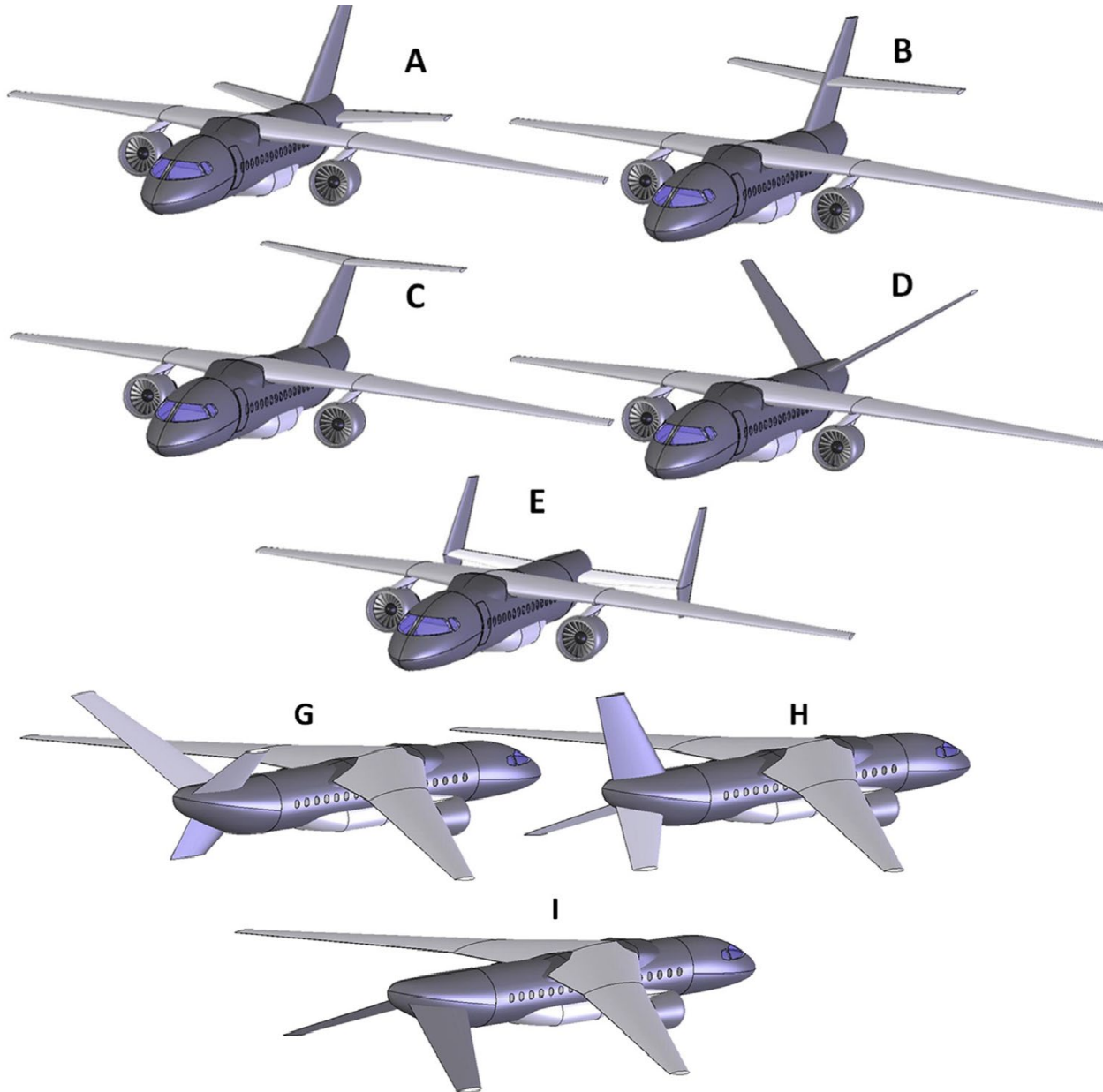
- ✈ Provide trim, stability and control



Longitudinal stability provided by horizontal tail

- ✈ Vertical tail required for unbalanced aerodynamic yawing moment.
 - Engine failure
 - Strong side winds during landing





A – Conventional

B – Cruciform

C – T-tail

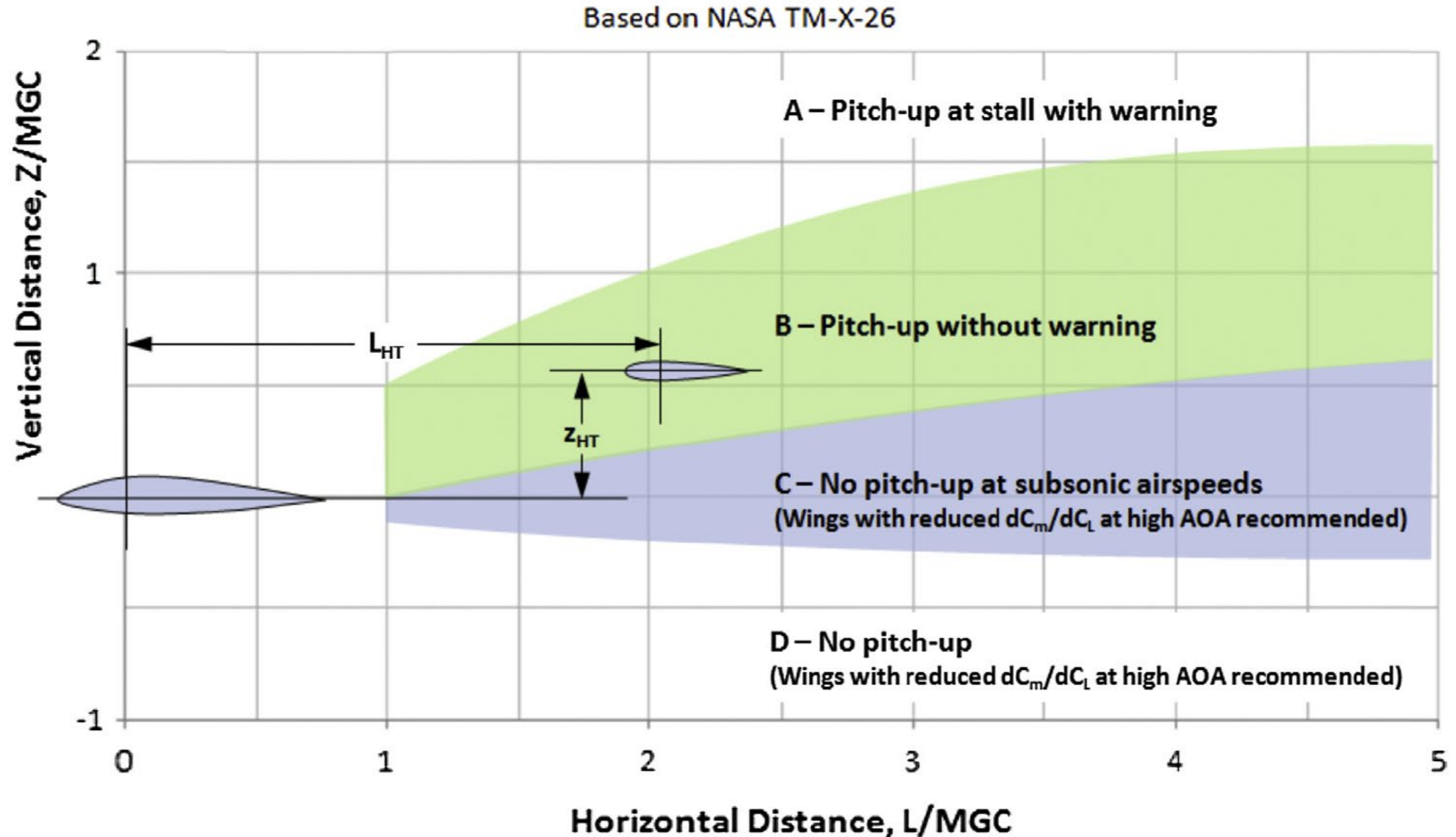
D – V-tail

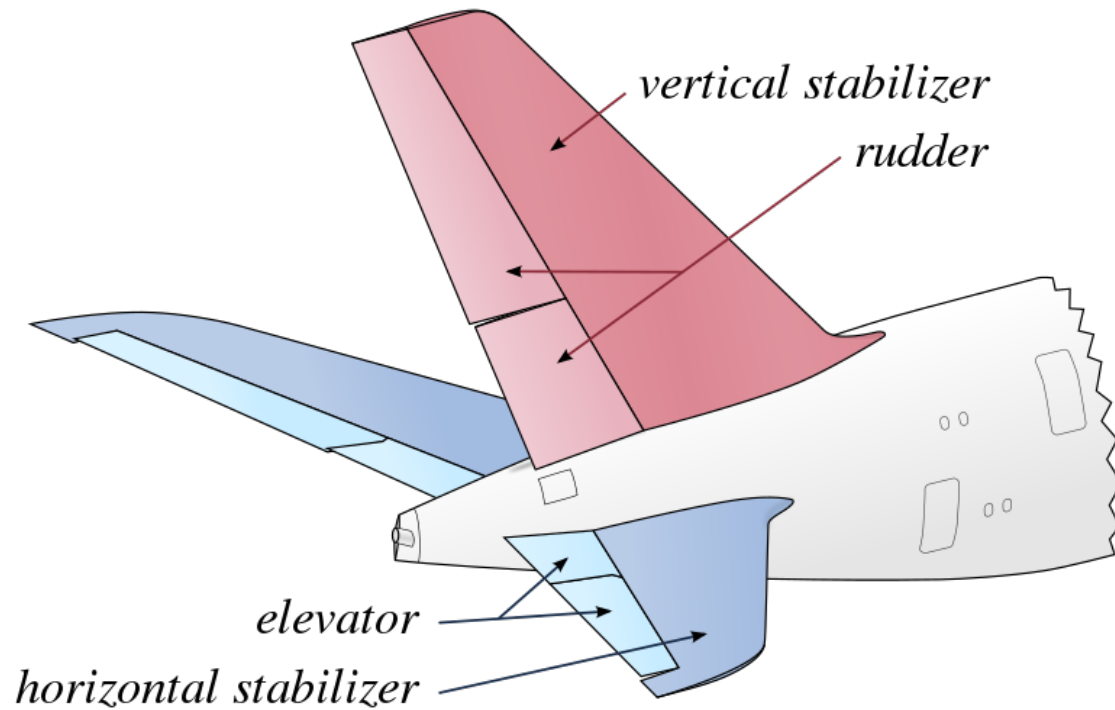
E – H-tail

G – Y-tail

H – Inverted Y-tail

I – Inverted V-tail





	Horizontal tail		Vertical tail	
	A	λ	A	λ
Jet fighter	3-4	0.2-0.4	0.6-1.4	0.2-0.4
Others	3-5	0.3-0.6	1.3-2.0	0.3-0.6
T-tail	3-5	0.3-0.6	0.7-1.2	0.6-1.0

Horizontal tail sweep:

- ✈ 5 deg more than wing sweep.
 - Delays stall
 - Increases critical Mach to avoid loss of effectiveness due to shocks

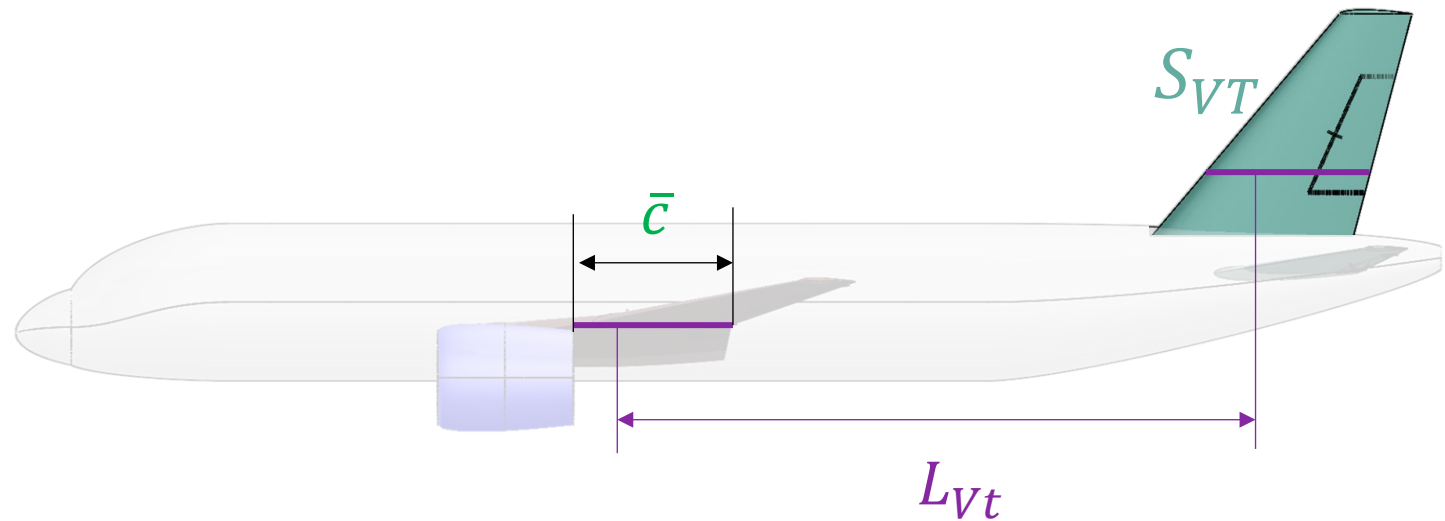
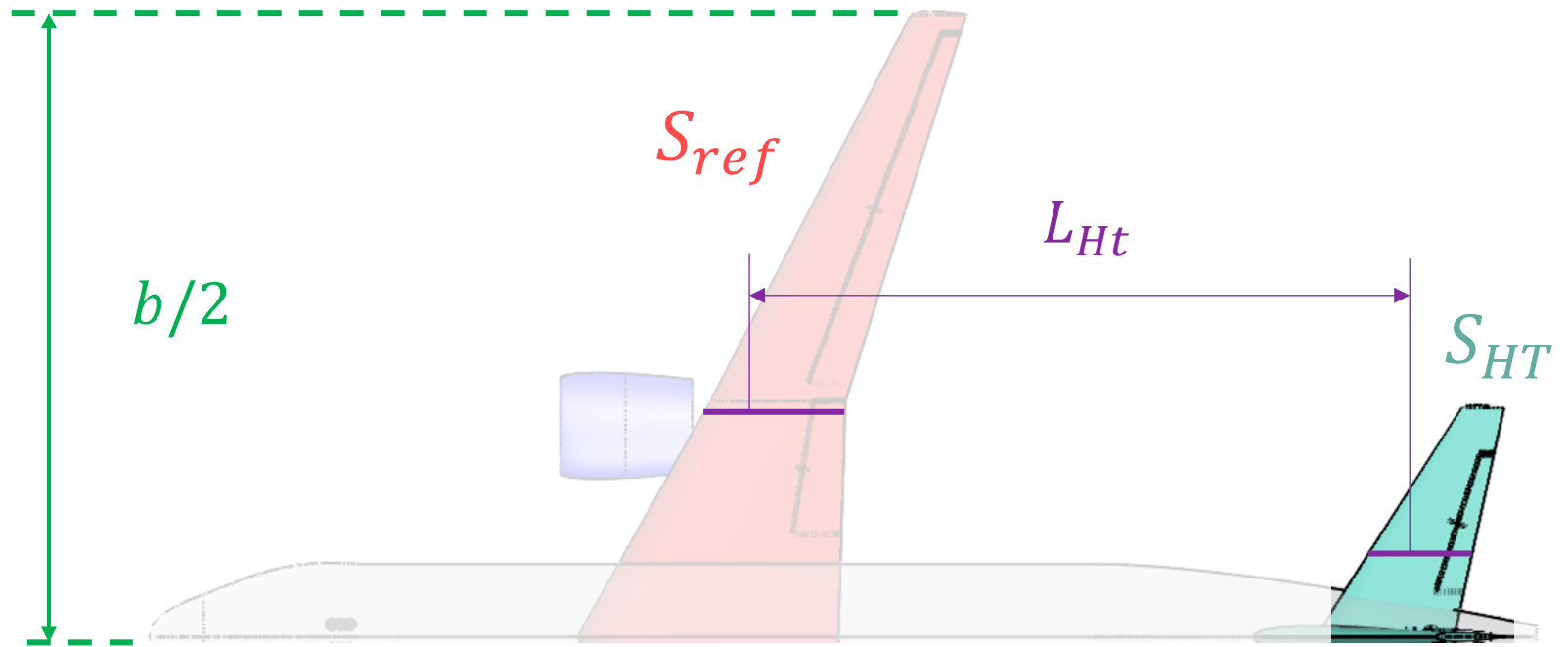
Vertical tail sweep:

- ✈ 35 to 55 deg

- ✈ Surface areas proportional to wing area:

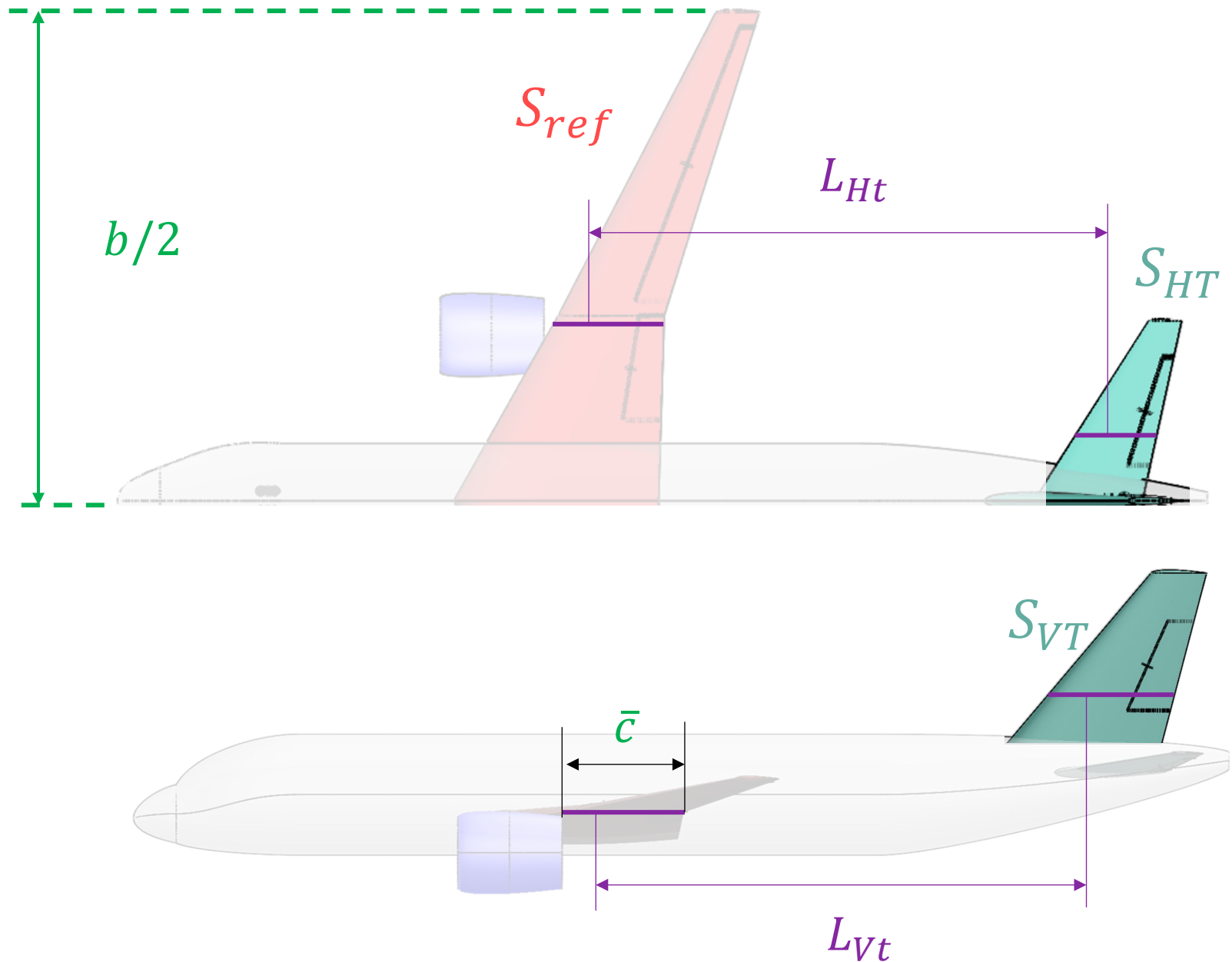
$$S_{VT} = \frac{c_{VT} b S_{ref}}{L_{VT}}$$

$$S_{HT} = \frac{c_{HT} \bar{c} S_{ref}}{L_{HT}}$$



- ✈ Surface areas proportional to wing area:

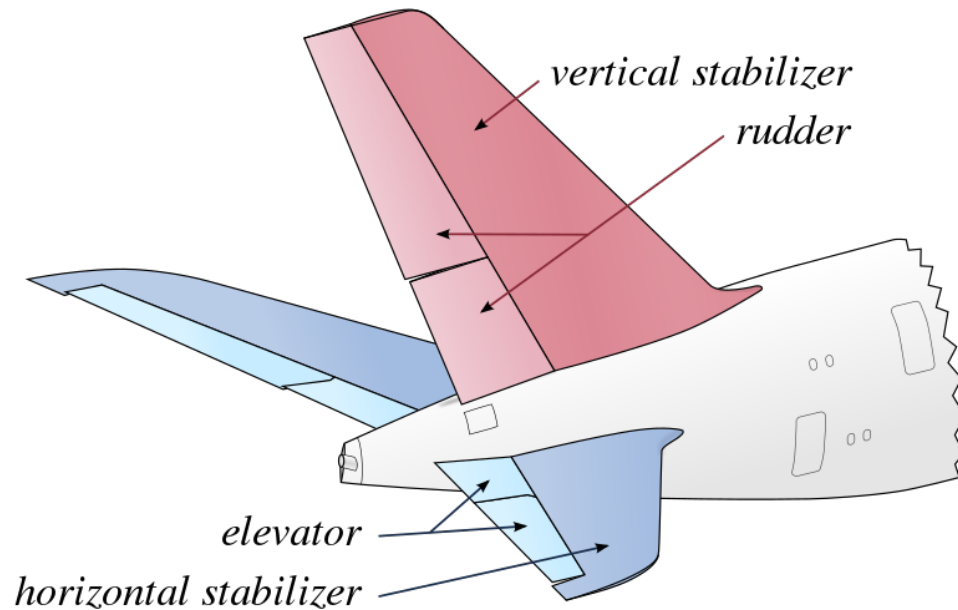
	c_{HT}	c_{VT}
Jet fighter	0.4	0.07-0.12
General aviation	0.7	0.04
Twin turboprop	0.9	0.08
Transport	1.0	0.08
Dash8 Q400	1.769	0.1252



	Elevator $\frac{C_e}{C}$	Rudder $\frac{C_r}{C}$
Jet fighter	0.3*	0.3
General aviation	0.45	0.4
Buisness jet	0.32 ⁺	0.3
Transport	0.25 ⁺	0.32

⁺ Typically all moving plus elevator

* Usually all moving without separated elevator



Thank You!



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